

WEBNOMAD STUDIO · WEBNOMAD.ORG

Retirement Income Planner

User Guide

Version 7

The first retirement planner you can chat with — that teaches you as you plan, verifies its own AI suggestions, and stays accurate for years.

Published: June 2026 · Supersedes the v6 User Guide
WebNomad Studio · webnomad.org · dev@webnomad.org

How to use this guide

This guide walks you through every feature of the Retirement Income Planner, version 7 — from opening the file for the first time to chatting with the new **Plan with AI** co-pilot and running AI audits of your help content. It assumes no prior knowledge of retirement-planning jargon: terms like MAGI, RMD, ACA cliff and IRMAA are explained the first time they appear.

You can read the guide front-to-back to learn the planner systematically, or jump to a chapter from the Table of Contents.

Two example users

Most chapters use one of two example users so screenshots tell a coherent story:

- **Linda** — a US single filer retiring at 62 with \$400,000 in her 401(k), \$50,000 in cash and \$80,000 in equity. She plans to claim Social Security at 67. Most US-tax, ACA and IRMAA chapters follow Linda's plan.
- **David & Jo** — a UK retired couple living in the Philippines on UK State Pension plus a small US Social Security entitlement. They appear in the multi-currency, foreign-resident and treaty chapters.

Conventions

- **Bold** marks button labels and tab names.
- *Italics* mark a phrase you'll see displayed in the planner.
- Numbered procedures (1, 2, 3...) are step-by-step actions — follow them in order.

Disclaimer

This guide and the Retirement Income Planner are educational tools for personal planning. They do not constitute financial, tax, legal or investment advice. All projections are estimates based on the values you enter — not guarantees. Tax law, ACA rules, Medicare premiums, Social Security structures and overseas tax parameters change frequently. The planner has tools to help you keep its figures and rules current (see Chapter 37), but you are responsible for verifying your plan reflects current law and your personal circumstances. Always consult a qualified financial advisor, tax professional or retirement planner before making any financial decisions.

What's new in version 7

Version 7 builds on v6's AI advisor, education system and longevity tools. If you are upgrading, here is everything that changed — every existing plan loads unchanged.

Area	What's new in v7	Chapter
Plan with AI co-pilot	A dockable AI sidebar you can open from any tab. It updates your plan in plain English, tests its own suggestions in the simulator before you apply them, auto-refines proposals that would run you out of money, and gives you one-click Undo .	20
Part-time income	Now available in every phase (was phases 1–2 only).	7
State income tax	Type-ahead state picker plus an AI auto-fill that looks up your state's flat rate and exemptions.	7, 10, 21
Enriched PDF report	Adds an asset-allocation breakdown, a plan-resilience page (Monte Carlo + the 12-scenario stress grid) and portfolio charts. Blank-page printing bug fixed.	23, 35
AI Chat	A copy icon on every AI response for pasting into Word or email.	19
Tabs & layout	Tabs reordered into a left-to-right workflow; a polished sticky header (lift shadow, frosted bar, sliding tab marker); grouped Overview toolbars.	33
Themes	Two new visual styles — Pro dashboard and High contrast (accessibility) — for five styles in total.	33
Education	New ⓘ popovers (getting-started, a distinct 22%-bracket explainer, a dedicated SS-claiming-age explainer, cash/bond volatility); survivor-scenario setup guidance.	25, 28
Feature Tour	Expanded from 7 to 9 stops — now includes Plan with AI and Report.	31
Historical backtest NEW	Replays your plan against the actual sequence of US market returns for every overlapping retirement window since 1928 — a sequence-of-returns confidence check alongside Monte Carlo. On the Stress test tab.	14
Lump sums — money out	Lump sums now work both ways: one-off expenses (roof, car, medical) as well as windfalls, with a per-row Money in / Money out selector.	8
Confidence summary NEW	Plan Health now opens with a plain-English summary uniting all three confidence lenses (checks, Monte Carlo, backtest) and a spending-headroom line.	11
Maximum sustainable spending NEW	A transparent "how much can I safely spend?" finder on the What-if? tab, with solvent / Monte Carlo / historical gates.	25
Education	New retirement spending smile concept (go-go / slow-go / no-go) with a Concepts-to-learn prompt and glossary entry.	30, 31

AI Assistant

Quick questions to get started:

Rebalance to grow equity

Optimise pre-Medicare for ACA

Fill the 12% bracket

Build a Roth conversion ladder

Optimise & set SS claim age

Smooth income across phases

Sidebar chat cleared. Use the quick questions below or ask anything.

Ask about your plan... (Enter to send)

AI Retirement Income Planner v7

Live rates ✓ Export JSON Import JSON Create Report Reset defaults ?

Scenario: Age 59.5, Downsize US Condo Phase 1 / 5-Phase Retirement Income Plan · ACA · CSR · Tax · Inflation-Adjusted Net Income · Lump Sums · Equities

USD GBP CAD AUD PHP THB Override rates

\$ plain Fit phases A A A A LIGHT STYLE Pro dashboard

OVERVIEW EDIT VALUES CASH LUMP SUMS BALANCE SCENARIOS STRESS TEST SS OPTIMIZER DRAWDOWN ANNUITIES WHAT-IF? TAX & ACA AI CHAT PLAN WITH AI

AVG NET / MO
\$3,680 nom - \$2,917 real
EST. TOTAL AGE 80 (USD)
\$89,142
PHASE 5 PROJECTION: Age 72-80 Extend to age 90 Total tax to 88: \$53,167
Plan Health: 8/9 Replan from today

STARTING 401K (USD) \$200,000 STARTING CASH (USD) \$20,000 STARTING EQUITY (USD) \$40,000

RETIREMENT PHASES

PHASE 2 Age 62-65 · 36mo

401k withdrawal \$1,900/mo
Cash withdrawal \$150/mo
Equity withdrawal \$350/mo
Gross income \$2,400/mo
Federal income tax -\$58/mo
ACA premium est. -\$79/mo
ACA subsidy eligible
ACA Silver CSR
10% bracket (\$12,962) ~2028 est.
Net income after tax & healthcare
Nominal \$2,263/mo
Real (today's money) \$2,165/mo
Deflated 3.5 yrs @ 3%
Ending balances
401k: \$178,718
Cash: \$16,671
Equity: \$39,384

PHASE 3 Age 65-67 · 24mo

401k withdrawal \$2,250/mo
Cash withdrawal \$150/mo
Equity withdrawal \$450/mo
Gross income \$2,850/mo
Federal income tax -\$73/mo
Medicare Part B est. -\$228/mo
Medicare Part D est. -\$53/mo
10% bracket (\$13,956) ~2030 est.
Net income after tax & healthcare
Nominal \$2,496/mo
Real (today's money) \$2,218/mo
Deflated 4.0 yrs @ 3%
Ending balances
401k: \$138,518
Cash: \$14,235
Equity: \$36,865

PHASE 4 Age 67-72 · 60mo

401k withdrawal \$1,350/mo
Cash withdrawal \$100/mo
Equity withdrawal \$450/mo
US Social Security \$1,949/mo
UK State Pension \$1,885/mo
Gross income \$5,654/mo
Federal income tax -\$356/mo
Medicare Part B est. -\$253/mo
Medicare Part D est. -\$59/mo
12% bracket (\$62,909) ~2034 est.
85% of SS taxable
Net income after tax & healthcare
Nominal \$4,986/mo
Real (today's money) \$3,995/mo
Deflated 7.5 yrs @ 3%
Ending balances
401k: \$99,785
Equity: \$24,491
Triple Lock: \$1,700 - \$1,805/mo
COLA: \$1,850 - \$1,949/mo

PHASE 5 Age 72-80 · 96mo

401k withdrawal \$880/mo
Cash withdrawal \$0 /mo
Equity withdrawal \$350/mo
US Social Security \$2,385/mo
UK State Pension \$2,191/mo
Gross income \$5,646/mo
Federal income tax -\$292/mo
Medicare Part B est. -\$307/mo
Medicare Part D est. -\$71/mo
12% bracket (\$76,235) ~2040 est.
75% of SS taxable
Est. 8940: \$303/mo
Net income after tax & healthcare
Nominal \$4,976/mo
Real (today's money) \$3,290/mo
Deflated 14.0 yrs @ 3%
Ending balances
401k: \$71,788
Equity: \$3,162
Triple Lock: \$1,700 - \$2,191/mo

Version 7: the Plan with AI co-pilot docks on the left while your dashboard stays fully visible on the right.

Table of contents

Part 1 — Getting started

1. What this planner does (and what it doesn't)
2. Opening the file for the first time
3. The Concepts Primer: 5 ideas worth knowing
4. The Setup Wizard: your first plan in 3 minutes
5. Already retired? The Replan-from-today path

Part 2 — Building your plan

6. Reading a phase card
7. The Edit tab: where the numbers live
8. Lump sums (money in and money out)
9. Currencies and residency modes
10. Tax parameters: what each one means

Part 3 — Understanding your results

11. Plan Health: the checks and the score
12. Acknowledging checks that don't apply
13. The five drawdown strategies
14. Stress Test (and the historical backtest)
15. Monte Carlo simulation
16. Scenarios comparison
17. Balance chart

Part 4 — AI features

18. Choosing an AI provider and adding your key
19. AI Chat: six starter prompts and how to read proposals
20. Plan with AI — the co-pilot sidebar **NEW**
21. Fetch current rates (and state-tax lookup)
22. Help content audit and one-click corrections
23. The PDF report (now with allocation, resilience & charts)

Part 5 — Specialized tools

24. Annuity Calculator
25. What-if Explorer (and the survivor scenario)
26. Social Security Optimizer
27. Roth Conversion Optimizer integration

Part 6 — Learn as you plan

28. ⓘ icons: click any field you don't understand
29. Clickable phase-card badges
30. Concepts-to-Learn panel
31. Glossary and the 9-step Feature Tour
32. Tax & ACA notes tab

Part 7 — Settings and maintenance

- 33. Text size, dark mode, themes & the polished header
- 34. Saving, exporting, importing
- 35. PDF export
- 36. Keeping your planner current
- 37. Troubleshooting and FAQ

Appendices

- A. Keyboard shortcuts
- B. JSON export field reference
- C. Where to learn more

Part 1 — Getting started

The planner is a single HTML file that runs entirely in your browser. There is no installation, no account, no subscription, and your plan stays on your computer unless you use the optional AI features. This part walks you through your first 15 minutes with the planner.

Chapter 1 — What this planner does (and what it doesn't)

What it does

The Retirement Income Planner models your retirement, month by month, across up to five distinct phases. You tell it what you have today (401k, cash, equity, Roth, pensions), what you'll draw in each phase, and what tax rules apply. It returns:

- Your net monthly income in every phase, in nominal and real (today's-money) terms.
- Your account balances at the start and end of each phase.
- Your effective tax rate, marginal bracket, ACA subsidy status, IRMAA risk and Required Minimum Distribution exposure.
- Plan Health checks (up to 12) and a 0–100 confidence score.
- Five drawdown-strategy comparisons plus a Monte Carlo simulation.
- A built-in AI advisor that knows your plan — available both as a full-screen **AI Chat** tab and, new in v7, as a dockable **Plan with AI** co-pilot sidebar that updates your plan in plain English and verifies its own suggestions before you apply them.
- A clean, multi-page **PDF report** — now including an asset-allocation breakdown, a plan-resilience page and portfolio charts.

What it doesn't do

- It is **not** financial advice. The math is accurate; what you do with it is your decision.
- It does not connect to your bank, brokerage or HMRC/IRS. You enter the numbers, and they stay on your computer.
- It does not predict markets. Returns and inflation are assumptions you set.
- It does not model every product. Variable annuities, complex options strategies and bespoke tax structures are out of scope.

Why a single HTML file?

Privacy: your plan stays on your machine unless you use the optional AI features. Longevity: an HTML file works in any browser, for years, without a hosted service. Portability: copy it to a USB stick, email it to your spouse, or share it with your advisor — it's just one file.

AI Retirement Income Planner v7

Scenario: Age 59.5, Downsize US Condo Phase 1 / 5-Phase Retirement Income Plan - ACA - CSR - Tax - Inflation-Adjusted Net Income - Lump Sums - Equities

Live rates ✓ Export JSON Import JSON Create Report Reset defaults ?

USD GBP CAD AUD PHP THB Override rates

\$ plain Fit phases \$ A A A LIGHT STYLE Pro dashboard

OVERVIEW EDIT VALUES CASH LUMP SUMS BALANCE SCENARIOS STRESS TEST SS OPTIMIZER DRAWDOWN ANNUITIES WHAT-IF? TAX & ACA AI CHAT PLAN WITH AI

AVG NET / MO \$3,680 nom \$2,917 real EST. TOTAL AGE 80 (USD) \$89,142 PHASE 5 PROJECTION: Age 72-80 Extend to age 90 Total tax to 88: \$83,167 Plan Health: 8/8 Replan from today

STARTING 401K (USD) \$200,000 STARTING CASH (USD) \$20,000 STARTING EQUITY (USD) \$40,000

RETIREMENT PHASES

PHASE 2

Age 62-65 - 36mo

401k withdrawal

\$1,900/mo

Cash withdrawal

\$150/mo

Equity withdrawal

\$350/mo

Gross income

\$2,400/mo

Federal income tax

ACA premium est.

✓ ACA subsidy eligible

✓ ACA Silver CSR

10% bracket (\$12,962) ~2028 est.

-\$58/mo

-\$79/mo

MAGI > 100% FPL

100-250% FPL

Net income after tax & healthcare

Nominal

\$2,263/mo

Actual dollars received

Real (today's money)

\$2,165/mo

Deflated 1.5 yrs @ 3%

Ending balances

401k: \$178,715

Cash: \$16,671

Equity: \$39,384

PHASE 3

Age 65-67 - 24mo

401k withdrawal

\$2,250/mo

Cash withdrawal

\$150/mo

Equity withdrawal

\$450/mo

Gross income

\$2,850/mo

Federal income tax

Medicare Part B est.

Medicare Part D est.

10% bracket (\$13,956) ~2030 est.

-\$73/mo

-\$228/mo

-\$53/mo

Net income after tax & healthcare

Nominal

\$2,496/mo

Actual dollars received

Real (today's money)

\$2,218/mo

Deflated 4.5 yrs @ 3%

Ending balances

401k: \$138,518

Cash: \$14,235

Equity: \$36,865

PHASE 4

Age 67-72 - 60mo

401k withdrawal

\$1,350/mo

Cash withdrawal

\$100/mo

Equity withdrawal

\$450/mo

US Social Security

\$1,949/mo

UK State Pension

\$1,805/mo

Gross income

\$5,654/mo

Federal income tax

Medicare Part B est.

Medicare Part D est.

≤12% bracket (\$62,909) ~2034 est.

85% of SS taxable

-\$356/mo

-\$253/mo

-\$59/mo

Net income after tax & healthcare

Nominal

\$4,986/mo

Actual dollars received

Real (today's money)

\$3,995/mo

Deflated 7.5 yrs @ 3%

Ending balances

401k: \$99,785

Cash: \$18,578

Equity: \$24,491

Triple Lock: \$1,700 - \$1,805/mo

COLA: \$1,850 - \$1,949/mo

PHASE 5

Age 72-80 - 96mo

401k withdrawal

\$800/mo

Cash withdrawal

\$0 /mo

Equity withdrawal

\$350/mo

US Social Security

\$2,385/mo

UK State Pension

\$2,191/mo

Gross income

\$5,646/mo

Federal income tax

Medicare Part B est.

Medicare Part D est.

≤12% bracket (\$76,235) ~2040 est.

75% of SS taxable

Est. RMD: \$303/mo

-\$292/mo

-\$307/mo

-\$71/mo

Net income after tax & healthcare

Nominal

\$4,976/mo

Actual dollars received

Real (today's money)

\$3,290/mo

Deflated 14.0 yrs @ 3%

Ending balances

401k: \$71,788

Cash: \$14,272

Equity: \$3,162

Triple Lock: \$1,700 - \$2,191/mo

COLA: \$1,850 - \$2,305/mo

The Overview dashboard: your whole retirement, phase by phase, at a glance.

Chapter 2 — Opening the file for the first time

You'll have received the planner as a download after purchase — a single .html file like Retirement-Planner-v7-v272.html. The number at the end is the build; it changes with each release, so yours may differ.

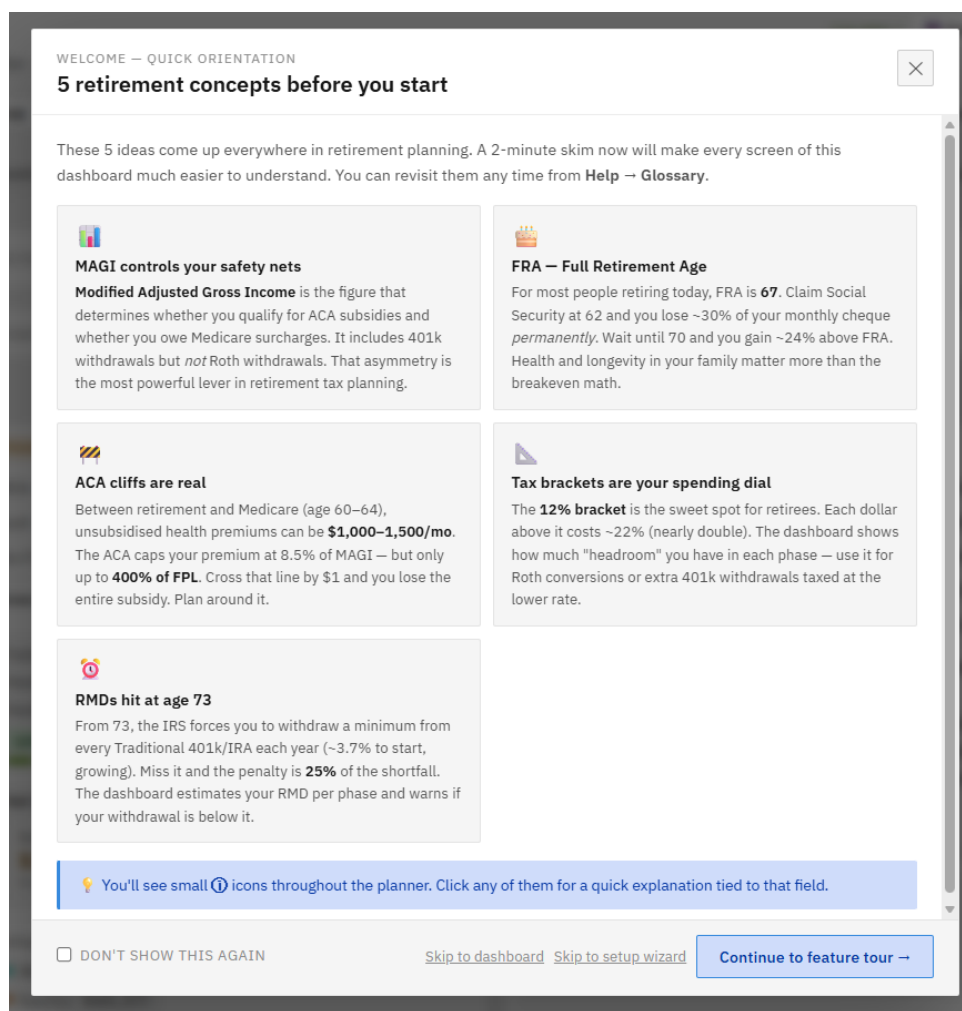
To open the planner:

1. Save the `.html` file to a folder on your computer (your Documents folder is fine).
2. Double-click the file. It opens in your default browser (Chrome, Edge, Firefox or Safari all work).

On first open, the **Concepts Primer** and **Setup Wizard** appear automatically (next two chapters).

Tip — make a desktop shortcut

Right-click the file and choose *Send to → Desktop (create shortcut)* on Windows, or drag it to your Dock on macOS. Your plan auto-saves to the browser's local storage on every change — re-open in the same browser and your plan is still there.



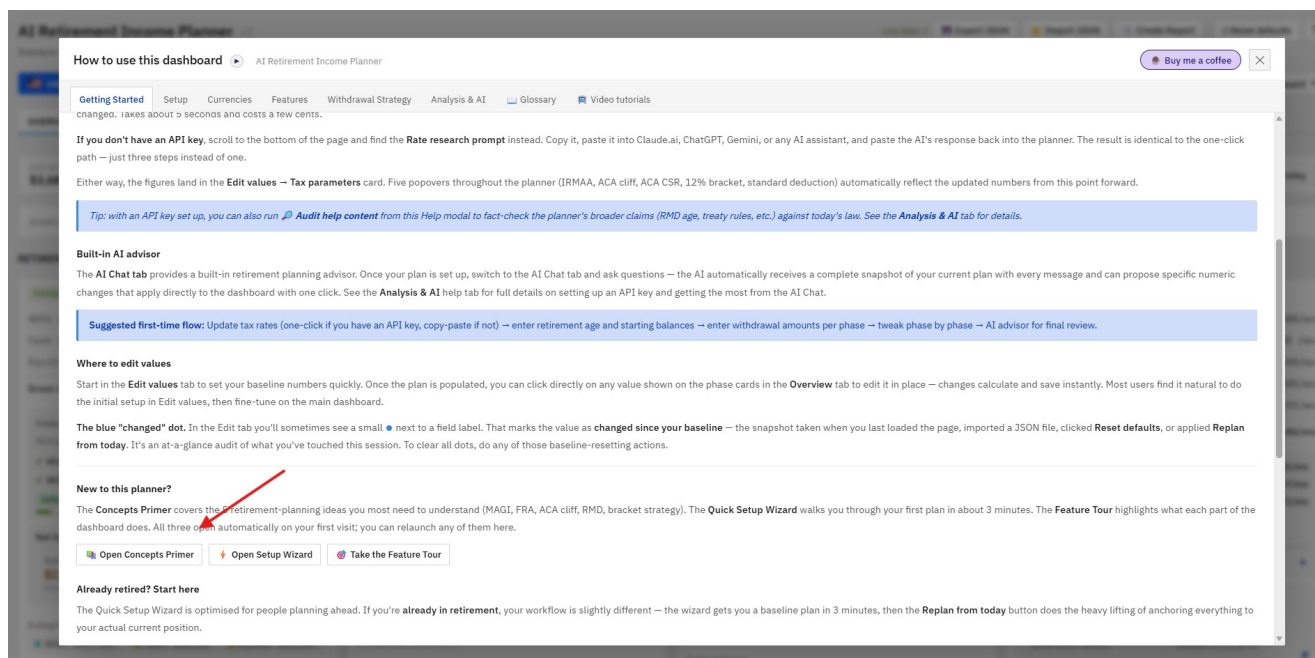
On first open, a five-card primer introduces the ideas that drive most retirement decisions.

Chapter 3 — The Concepts Primer: 5 ideas worth knowing

Before you build your first plan, the planner shows a short primer covering five ideas that drive most decisions. Skip it on your first visit if you like — you can relaunch it any time from the Help modal.

The five concepts

- 1. MAGI controls your safety nets.** Modified Adjusted Gross Income determines whether you qualify for ACA subsidies and whether you owe Medicare surcharges (IRMAA). It includes 401k withdrawals but *not* Roth withdrawals — that asymmetry is the most powerful lever in retirement tax planning.
- 2. FRA — Full Retirement Age.** For most people retiring today, FRA is 67. Claim at 62 and you lose about 30% permanently; wait to 70 and you gain about 24% above FRA.
- 3. ACA cliffs are real.** Between retirement and Medicare, unsubsidized premiums can be \$1,000–1,500/month. The ACA caps your premium at a share of MAGI that rises with income — but only up to 400% of the Federal Poverty Level. Cross by \$1 and you lose the entire subsidy.
- 4. Tax brackets are your spending dial.** The 12% bracket is the sweet spot; each dollar above it costs about 22%. The planner shows your *headroom* in each phase.
- 5. RMDs hit at 73 or 75.** From 73 (75 if born in 1960 or later) the IRS forces a yearly minimum withdrawal from Traditional 401k/IRA. Miss it and the penalty is 25% of the shortfall.



Reopen the primer any time from Help → Getting Started.

Chapter 4 — The Setup Wizard: your first plan in 3 minutes

After the Primer (or any time from the Help modal), the Setup Wizard guides you through the inputs a working plan needs. You can skip any step and fill it in later on the Edit values tab. Currency and residence are set with the buttons along the top of the dashboard (Chapter 9).

The wizard's six steps

- 1. Welcome.** A short tour of what the planner does.
- 2. Your starting numbers.** Your retirement start age, then your 401k / IRA, cash, Roth and brokerage balances and their growth rates. If you are still earning when the year you retire begins, enter that year's salary and the month you stop: it is taxed with that year's withdrawals and counts toward that year's ACA subsidy. Retiring within about two years of 65, you are also asked for your income in each of the two years before you retire, because Medicare reads those tax returns for your first premiums.
- 3. Monthly income sources.** Social Security at your claiming age with its COLA and your birth year, any UK State Pension, your inflation assumption, a US pension or disability income, and your first monthly withdrawals from the 401k and cash.
- 4. Tax & settings.** Filing status, any rental or passive income, the RMD start age (leave it blank and it follows your birth year: 73, or 75 if you were born in 1960 or later) and your state income-tax rate. The federal figures ship with the planner; with an AI provider set up, **Fetch current rates** updates them (Chapter 21).
- 5. Your goals (optional).** A target monthly income in today's money and a legacy balance, which Plan Health then checks.
- 6. You're ready to plan!** A short summary. Your plan runs to age 90 unless you change the Age control on the tab bar.

After the wizard

You'll land on the Overview dashboard. Fine-tune any input on the **Edit values** tab or by clicking values inline on the phase cards. The Plan Health badge immediately shows whether your plan passes its checks — don't worry if not every check passes (see Chapters 11–12).

⚡ Quick Setup Wizard



1

2

3

4

5

6

Your starting numbers

Enter your current savings balances and retirement age. All values in USD — use the currency buttons to display in GBP, EUR, PHP, or THB later.

Retirement start age

62

Earliest penalty-free 401k access: 59½. **Already retired?** Enter the age you actually retired at — you'll use the *Replan* button after the wizard to fast-forward to your current age.

Still earning a salary the year you retire? It is taxed together with that year's withdrawals and counts toward that year's ACA subsidy. Enter it below.

Salary earned in the year you retire, before you stop (USD)

30000

Month you retire

April



Only that year's pay before you stop, not a full year's salary (retiring at the end of June on \$150,000 a year: about \$75,000). Taxed together with that year's withdrawals, and it counts toward that year's ACA subsidy.

401k / IRA balance (USD) [▶ Traditional 401k · 2 min](#)

400000

Tax-deferred — every withdrawal counts as ordinary income

Cash / savings balance (USD)

50000

Post-tax, no income tax on withdrawals

Roth IRA balance (USD)

[▶ Roth mechanics · 2 min](#)

0

Tax-free growth & withdrawals

Brokerage / taxable investments balance (USD)

80000

Stocks and funds held outside retirement accounts. Only the gain in what you sell is taxed — set how much of the balance is gain on the Edit tab. Enter 0 if you have none.

The wizard's second step: your balances and, if you are still earning that year, the salary in the year you retire.

Chapter 5 — Already retired? The Replan-from-today path

Most planners assume you're modeling forward from accumulation. If you're already retired, that's the wrong starting point. **Replan from today** is designed for you.

How it differs from the normal wizard

The wizard asks *"at what age will you retire?"* Replan-from-today asks *"how old are you now, and what are your actual current balances?"* It then:

- Trims past phases out of the model so they don't double-count.
- Pre-fills your current Social Security and UK State Pension amounts, and moves your spouse's age on by the same number of years if you file jointly.
- Restarts the projection from your actual current balances.
- Asks for your income in each of the last two years, which Medicare reads for your next premiums (it starts at the plan's own projection), and remembers recent Roth conversions for the five-year seasoning check.

Recommended flow

1. Step through the Setup Wizard with your real retirement start age.
2. On the Edit values tab, click **Fetch current rates** (or use the **Rate Research Prompt**).
3. Click **Replan from today** and enter your current age and actual balances.
4. Open the **Plan with AI** sidebar (Chapter 20) and ask for optimizations from your anchored position.

AVG NET / MO

\$4,112

nom

\$3,230

real

EST. TOTAL AGE 80 (USD)

\$1,090,697

PHASE 5 PROJECTION:

Age 72-80

Extend to age 90

Total tax to 80: \$60,839

Plan Health: 7/8 ✓

Replan from today

REPLAN FROM CURRENT AGE

Replan - 90 sec

First time using Replan? Enter today's reality — your actual current age, current account balances, and current SS / UKP payment amounts (the amounts you receive now, after any COLA / triple-lock growth that has already happened). Phases you've already lived through will be trimmed and the rest of the plan re-anchored from your real situation. Your original plan stays saved until you Apply — and you can roll back any time with Reset to original plan.

CURRENT AGE

p.g. 63

ACTUAL 401K BALANCE (USD)

400000

ACTUAL CASH BALANCE (USD)

50000

ACTUAL EQUITY BALANCE (USD)

80000

✓ Apply & recalculate

Cancel

↔ Reset to original plan

Replan from today anchors an already-retired plan to your real current age and balances.

Part 2 — Building your plan

With your first plan loaded, this part walks through the day-to-day mechanics of inspecting and editing your numbers.

Chapter 6 — Reading a phase card

The **Overview** tab shows your retirement as a strip of phase cards (typically five). For Linda these are roughly ages 62–65, 65–67, 67–72, 72–80, and 80+.

Anatomy of a phase card

- **Age range and label** — e.g. *Phase 2 (ages 65–67)*.
- **Net monthly income** — what you'll receive after tax, in nominal and real terms.
- **Income source breakdown** — color-coded segments for 401k, cash, equity, Roth, Social Security, pensions, part-time work and rental income.
- **Tax & healthcare bar** — federal tax, ACA premium or Medicare cost, and IRMAA surcharge.
- **Account balances** — start and end balance per account with direction arrows.
- **Health badges** — clickable badges flagging conditions like *Near 12% bracket ceiling* or *Below RMD estimate*.

New in v7 — clearer phase cards

The "*% of SS taxable*" note now always sits on its own line beneath the bracket badge, so it never crowds the bracket pill at narrow widths. The Roth withdrawal and conversion sections now show each phase's duration in months (e.g. *Phase 5 · 96mo*) so you can see the period a figure covers at a glance.

Clickable badges

Every badge is clickable, and the explanation uses your actual figures — for instance: "*You're \$4,200 below the 12% ceiling — you could withdraw an extra \$4,200 from your 401k or convert that much to Roth without bumping into 22%.*"

PHASE 2 Age 62–65 · 36mo

401k withdrawal \$1,900/mo

Cash withdrawal \$150/mo

Equity withdrawal \$350/mo

Gross income \$2,400/mo

Federal income tax -\$58/mo

ACA premium est. -\$79/mo

✓ ACA subsidy eligible **MAGI > 100% FPL**

✓ ACA Silver CSR **100–250% FPL**

10% bracket (\$12,962) ~2028 est.

Net income after tax & healthcare

Nominal

\$2,263/mo

Actual dollars received

Real (today's money)

\$2,165/mo

Deflated 1.5 yrs @ 3%

Ending balances

401k: \$417,303

Cash: \$50,238

Equity: \$93,231

Every element of a phase card — and every badge is clickable for an explanation in your own numbers.

Chapter 7 — The Edit values tab: where the numbers live

Click the **Edit values** tab to see every input that drives your plan. The buttons along its top show one group of cards at a time.

The six sections

- **Accounts** — starting balances and growth rates, equities (with their pre-plan gain), the Roth IRA and your goals.
- **Income** — Social Security and pensions (your spouse's too, if married), the survivor scenario, your US tax obligation and filing status, rental or passive income, Canadian and Australian pensions, and inflation.
- **Phases** — the phase age boundaries and a card for each phase: what you withdraw from each account, plus part-time income and Roth conversions.
- **Taxes & healthcare** — healthcare costs and the IRMAA options, your income around retirement (below), and the US, married and UK tax parameters (Chapter 10).
- **Advanced** — RMD settings (a blank start age follows your birth year) and Monte Carlo settings, including the **Backtest growth-bucket stock %** (default 60/40) used by the Historical backtest (Chapter 14).
- **Appearance** — the colors used for amounts.

New — your income around retirement

Taxes & healthcare asks for the salary you earn in the year you retire and the month you stop: it is taxed with that year's withdrawals and counts toward that year's ACA subsidy. Retiring near 65, it also asks for your income in the two years before, which Medicare reads for your first premiums. Tick SSA-44 if you will ask Social Security to use your lower retirement income instead.

New in v7 — state income tax with autocomplete + AI lookup

The **State** field offers a type-ahead list of US states as you type. Once a state is entered, an **Auto-fill rate** button becomes active: click it and the AI looks up that state's flat-rate approximation and any retirement-income exemptions and fills them in for you. Clear the state and the button disables again. (Requires an API key — see Chapter 18.)

Inline editing & blue-dot indicators

You can change any number anywhere. Edit-tab inputs update the dashboard live, and you can click directly on dashboard phase-card numbers to edit in place (USD display mode only). A small blue dot marks any field changed since the session started; save or import a plan to reset the baseline.

PHASE 3 Age 65–67 · 24mo

401k withdrawal	\$2,250/mo
Cash withdrawal	\$150/mo
Equity withdrawal	\$450/mo
Part-time work	\$1,200/yr
Gross income	\$2,950/mo

Federal income tax	–\$83/mo
Medicare Part B est.	–\$228/mo
Medicare Part D est.	–\$53/mo

10% bracket (\$13,956) ● ~2030 est.

Net income after tax & healthcare

Nominal
\$2,586/mo
Actual dollars received

Real (today's money)
\$2,298/mo
Deflated 4.0 yrs @ 3%

Ending balances

● 401k: \$422,035 ● Cash: \$50,412
● Equity: \$101,877

A phase card with part-time work; the blue dot marks a value you changed this session.

Chapter 8 — Lump sums (money in and money out)

Real retirements include one-time cash events — selling a house, an inheritance, a deferred bonus, but also one-off expenses like a new roof, a car, a wedding, or a big medical year. The **Lump sums** tab lets you add both kinds without distorting your phase withdrawal rates.

Adding a cash event

1. Click the **Lump sums** tab, then **+ Add cash event**.
2. Set its **Type** — **Money in** (a windfall) or **Money out** (a one-off expense).
3. Enter the amount, the phase it lands in, and an optional label (e.g. House sale or New roof).
4. Choose the accounts in the **Paid from** / **Paid into** column.

For an expense, **Smart order (cash first)** takes cash, then the brokerage account, the 401k, the Roth and Super. Pick one account instead and only that account pays; anything it cannot cover shows as unfunded unless you tick **Use other accounts if short**. **Custom split** sets how much each account gives or receives. Money in lands in cash unless you choose another account.

Money in adds funds — it doesn't spend them

A *money-in* lump increases your cash balance but doesn't automatically increase what you spend. To use it for living expenses, raise your cash withdrawal in the same or a later phase on the Edit values tab.

Tax on a one-time expense

Money taken from a 401k is taxed as income that year (with the 10% penalty before 59½), and selling brokerage holdings taxes the gain, so less than you withdraw reaches the expense. Tick Withdraw extra to cover the tax to gross it up so the full amount arrives. Cash and Roth money is not taxed.

Taxable money coming in

Tick **Taxable income in the year it arrives** for a pension commutation, a deferred bonus or an annuity surrender's gain. It is taxed once, as that year's income; any part you pay into the 401k / IRA is taxed only as you withdraw it.

The year it lands

An event's income counts in that year: in the ACA subsidy, in the IRMAA Medicare reads two years later, in the 3.8% Net Investment Income Tax, and in how much of your Social Security is taxable. The line under each event shows what it costs.

On the phase cards, money in shows as a green "+ \$X lump sum added" pill and money out as a neutral "– \$X one-time expense" pill; an amber pill warns when an event crosses the ACA cliff or an IRMAA line that year. For money-in rows, an **Evaluate as annuity** link opens the Annuity Calculator with the lump as the premium (Chapter 24).

● Lump sum events ⓘ

Add one-time cash events — **money in** (a windfall such as a house sale or inheritance) or **money out** (a one-off expense such as a new roof, car, or big medical year). Enter amounts in **USD**; each is applied at the start of its assigned phase. Use *Paid from / Paid into* to choose the accounts involved — pick one account and the planner leaves the rest alone, showing you any shortfall rather than quietly selling something else to cover it.

1 MONEY OUT · ONE-TIME EXPENSE · NEW ROOF

Type	Amount (USD)	Phase	Label	Paid from
Money out ▼	40000	Phase 2 (62–65) ▼	New roof	401k / IRA ▼ ×

- ☐ Use other accounts if short
☒ Withdraw extra to cover the tax

Takes \$43,065 — \$43,065 from the 401k / IRA · **\$3,065 goes to tax** · **\$40,000 reaches the expense**.

Grossed up: the withdrawal is \$3,065 larger so the full \$40,000 still gets there after tax.

2 MONEY IN · WINDFALL · PENSION LUMP SUM

Type	Amount (USD)	Phase	Label	Paid into
Money in ▼	100000	Phase 3 (65–67) ▼	Pension lump sum	Custom split... ▼ ×

PUT INTO EACH ACCOUNT (USD)

Cash / savings	40000
Taxable brokerage	0
401k / IRA	60000
Roth	0

Fully allocated — \$100,000 of \$100,000.

- ☒ Taxable income in the year it arrives

Adds \$100,000 — \$40,000 into cash / savings, \$60,000 into the 401k / IRA.

The \$60,000 paid into the 401k is taxed as you withdraw it, not now; the rest is taxed in the year it arrives (about \$2,269).

[🔥 Evaluate as annuity →](#)

[+ Add cash event](#)

Each event chooses its accounts, and the line under it shows what it costs.

Chapter 9 — Currencies and residency modes

The planner supports seven currencies and four residency modes. The combination determines which tax regimes apply and which health-care assumptions are used.

Switching currency & the two currency flags

Use the currency selector in the header (USD / GBP / CAD / AUD / PHP / THB / EUR). All values are stored internally in USD; the display converts using live exchange rates. Two flags control behavior: *active currency* (your primary residency currency) and *display currency* (what's shown). You can browse a plan in one currency while modeling it in another.

Residency modes

- **US resident (USD)** — full US federal tax, IRMAA, ACA subsidy and Medicare modeling.
- **UK resident (GBP)** — UK income tax on 401k withdrawals, State Pension, equity gains and part-time income; US Social Security is taxed in the UK, not the US, under Article 17(3) of the US-UK treaty, which makes social security taxable only in the country you live in. US tax is reduced by a Foreign Tax Credit.
- **Foreign resident (PHP / THB / EUR / other)** — disables US healthcare costs (ACA, Medicare, IRMAA). US federal tax still applies if you remain a US person.
- **Canada / Australia** — full Canadian or Australian tax regime including CPP/OAS or Super/Age Pension.

New in v7 — close the custom-rates panel without losing your rates

If you enter your own exchange rates, you can now click **Done** to close the custom-rates panel while *keeping* those rates active — the reopen control is relabeled *Edit custom rates*. A separate *Use live rates* button discards your custom values and refetches live rates.

AI Retirement Income Planner v7
Scenario: Age 59.5, Downsize US Condo Phase 1 / 5-Phase Retirement Income Plan · ACA · CSR · Tax · Inflation-Adjusted Net Income · Lump Sums · Equities

USD

GBP

CAD

AUD

PHP

THB

Exchange rates (1 USD = ?)
Custom rates — edit below, changes apply instantly

1 USD = ? GBP
0.7433

1 USD = ? CAD
1.3833

1 USD = ? AUD
1.3957

1 USD = ? PHP
61.8053

1 USD = ? THB
32.6025

↻ Use live rates

Done

Set your own exchange rates, then click Done to keep them active while you keep planning.

Chapter 10 — Tax parameters: what each one means

On the Edit values tab, the **Taxes & healthcare** section holds the values that drive every tax calculation. Brackets and deductions are inflated forward to each phase's midpoint; the ACA's poverty lines and the IRMAA brackets are re-set year by year.

The federal parameters (US)

- **Standard deduction** — single and married, plus the extra deduction at 65 (in the phase you turn 65 it counts from your birthday). The temporary 2025–2028 senior deduction is not included.
- **Tax bracket ceilings** — for the 10%, 12%, 22%, 24%, 32% and 35% brackets; income above the last is taxed at 37%.
- **IRMAA threshold** — Medicare reads the tax return from two calendar years earlier, so each month's premium is priced on that year's income and brackets. **Two years back, Assume relief granted** (SSA-44) or **Premium from my notice** sets which income it reads, or replaces it with the premium your notice quotes.
- **FPL 100% / 250% / 400%** — the poverty levels behind ACA subsidies; each year of a phase is priced on its own line.
- **NIIT threshold** — where the 3.8% Net Investment Income Tax starts (\$200,000 single, \$250,000 married; fixed in law, not inflated).
- **State income tax** — a flat-rate approximation plus exemptions for your state, with type-ahead entry and optional AI auto-fill (Chapter 7).

UK / Canadian / Australian parameters

UK: personal allowance, 20% and 40% ceilings, 45% additional-rate threshold (USD-equivalent, auto-converted). Canadian: basic personal amount, four bracket ceilings plus average provincial rate. Australian: tax-free threshold, three bracket ceilings, the four marginal rates themselves, and the Medicare levy with its income threshold.

Keeping these current

Use **Fetch current rates** (Chapter 21) for one-click updates with an API key, or the **Rate Research Prompt** for any external AI. See Chapter 36 for the long-term workflow.

TAX PARAMETERS (2026 USD)

Current-year US federal thresholds. The dashboard automatically inflates these for each future phase using your inflation rate. Click any ⓘ for a plain-English explanation of why each field matters. To refresh these values for the current tax year, use the **Rate research prompt** at the bottom of this page.

STANDARD DEDUCTION — SINGLE (\$) ⓘ

16100

AGE 65+ ADDITIONAL DEDUCTION (\$) ⓘ

2050

10% BRACKET CEILING (TAXABLE INCOME \$) ⓘ

12400

12% BRACKET CEILING (TAXABLE INCOME \$) ⓘ

50400

22% BRACKET CEILING (TAXABLE INCOME \$) ⓘ

105700

IRMAA THRESHOLD — SINGLE (\$) ⓘ

109000

NIIT MAGI THRESHOLD — SINGLE (\$) ⓘ

200000

3.8% surtax on investment income above this MAGI (MFJ uses \$250k). Not inflation-indexed in law, but the model inflates it per phase like other thresholds.

ACA 400% FPL CLIFF — SINGLE (\$) ⓘ

62600

ACA 250% FPL — SILVER CSR CEILING (\$) ⓘ

39125

FEDERAL POVERTY LEVEL 100% — SINGLE (\$) ⓘ

15060

Every tax parameter in one place — inflation-adjusted to each phase automatically.

Part 3 — Understanding your results

With your plan built, this part covers how to interpret the planner's outputs — from the Plan Health score to the drawdown strategies and the Monte Carlo fan chart.

Chapter 11 — Plan Health: the checks and the score

Plan Health is a quick-glance score, shown as a badge on the dashboard (e.g. 9/11). Click it to open the full panel.

The checks, by tier

1. **Essential** — Income Adequacy (income stays positive and real purchasing power holds above half its starting level) and Portfolio Survives to End (balances last through your final phase).
 2. **Recommended** — Stress Test Resilient (the accounts you draw from survive returns 3 points lower with 5% inflation), Income Floor Foundation, RMD Compliance, Survivor Income Resilience, One-Time Expenses Funded and Roth Conversion Seasoning.
 3. **Optimization** — Tax Bracket Efficiency, ACA Subsidy Protected (pre-Medicare phases stay below the 400% FPL cliff), CSR / Subsidy Headroom and IRMAA Not Triggered. These flag money left on the table, not a failing plan.
 4. **Optional** — Income Goal and Legacy Goal, which count only when you set those goals.
 5. **Not counted** — checks that don't apply to your plan. Survivor Income Resilience needs a married plan with the survivor scenario on; the tax, ACA, IRMAA, RMD and Roth checks count only for US plans (IRMAA not with a premium from your notice); One-Time Expenses Funded, Roth Conversion Seasoning and CSR / Subsidy Headroom appear only when your plan has what they test.
- The badge color reflects the percentage passed: green $\geq 75\%$, amber $\geq 50\%$, red below. Each check has an expandable **Why this matters** explanation. Acknowledged checks add (*N acknowledged*) to the badge.

The Confidence summary NEW

At the top of the Plan Health panel, a short **Confidence summary** draws your three independent confidence lenses into one or two plain-English sentences — for example: *"Your plan passes 5 of 8 Plan Health checks (1 warning). In Monte Carlo it stays solvent in 92% of randomized runs to age 90. Against real market history it survived 78 of 78 retirements since 1928 (100%); the toughest start year was 1929."* A closing line names the **main thing to watch** — your highest-impact failing check if you have one, otherwise the residual risk (typically a poor first decade of returns). It also shows a **spending-headroom** line — roughly how much your plan sustains before the drawn-from accounts would run dry — linking through to the What-if Maximum-sustainable-spending tool (Chapter 25).

Why three lenses matter

The summary deliberately combines a *checklist* (Plan Health), a *random simulation* (Monte Carlo) and a *replay of real history* (the backtest, Chapter 14). When all three point the same way, that agreement is a stronger signal of a robust plan than any single number. The Monte Carlo line appears only once you've run that simulation (Drawdown Strategies → Monte Carlo); until then the summary invites you to add it. An ✨ **Ask AI about this** button hands the whole summary to the AI and asks for the single most impactful change.

✕ PLAN CONFIDENCE SCORE



▶ Plan Health Score · 3 min

Overall Plan Score

7/8 — Strong Plan



7 of 8 checks passed · 1 warning

Not every check needs to pass. Use the flags to find the highest-impact changes, then stop when you feel confident — or dismiss checks that don't apply to your situation.

Confidence summary

Your plan passes **7 of 8** Plan Health checks (1 warning). Against real market history it survived **81 of 81** retirements since 1928 (**100%**); the toughest start year was 1930.

Main thing to watch: **Stress Test Resilient.**

Spending headroom: your plan sustains about **\$5,741/mo** before the drawn-from accounts would run dry (your plan averages \$5,408/mo) — see **What-if? → Maximum sustainable spending.**

Open **Drawdown Strategies → Monte Carlo** to add the simulation lens to this summary.

Three independent lenses — a checklist, random simulation, and real history — pointing the same way is what confidence looks like.

🔥 [Ask AI about this](#)

Plan Health checks plus a plain-English Confidence summary at the top — each check with a personalized status and a Why-this-matters explanation.

PLAN HEALTH — 8 CHECKS



Income Adequacy

All phases show positive income and real purchasing power remains above 50% of the starting phase. ✓

▶ [Why this matters](#)



Tax Bracket Efficiency

All phases keep taxable income within efficient bracket territory. ✓

▶ [Why this matters](#)



ACA Subsidy Protected

ACA subsidies are protected across all pre-Medicare phases — MAGI sits between 100% and 400% FPL. ✓

Chapter 12 — Acknowledging checks that don't apply

Some checks won't be relevant — an ACA cliff check is irrelevant for a foreign-resident plan; an RMD check is irrelevant if your plan ends before 73. Rather than living with a permanent red flag, acknowledge the check.

1. Open the Plan Health panel.
2. Find the check you want to acknowledge.
3. Click **I'm OK with this** beneath its description.
4. Optionally enter a short reason (for your own records).
5. Click **Acknowledge**.

The check stays in the panel but faded, with a **Restore** link. Acknowledgments persist across reloads and travel with your plan when you export JSON.



ACA Subsidy Protected

ACA subsidies are protected across all pre-Medicare phases — MAGI sits between 100% and 400% FPL. ✓

▶ [Why this matters](#)



IRMAA Not Triggered

MAGI stays below IRMAA thresholds in all phases — no Medicare surcharges triggered. ✓

▶ [Why this matters](#)



Portfolio Survives to End

Portfolio balances remain healthy through all phases. ✓

▶ [Why this matters](#)



RMD Compliance

RMD shortfall of \$726/mo. IRS penalties (25% of shortfall) apply. Increase 401k withdrawals in the RMD phase immediately.

I'm OK with this ▶ [Hide explanation](#)

From age 73 the IRS requires you to withdraw a minimum amount from every Traditional 401k/IRA each year. The check flags any shortfall.

Why it matters: the penalty for missing an RMD is **25% of the shortfall** (10% if corrected within two years). Most people don't know about RMDs until the IRS reminds them.

▶ RMD • 2 min

▶ Roth conversion • 3 min



Income Floor Foundation

Guaranteed income covers 105% of average monthly income — a strong, resilient floor that protects the majority of your income from market risk. ✓

▶ [Why this matters](#)



Stress Test Resilient

Drawn-from accounts survive a bear scenario (returns -3%, inflation 5%) to age 80. The plan has meaningful resilience against adverse conditions. (Untouched legacy balances are

Consciously dismiss a check that doesn't apply — with an optional note for your records.

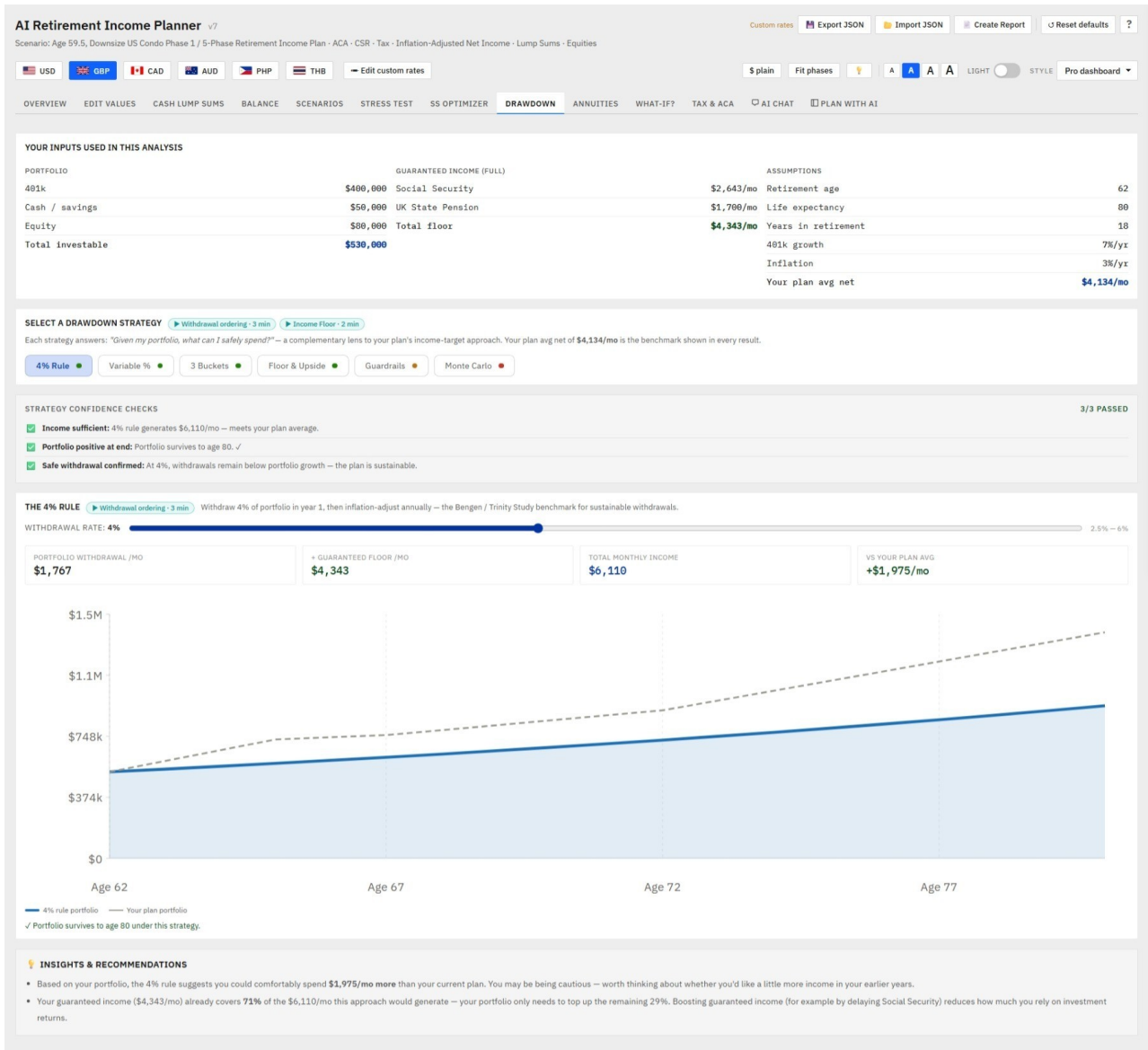
Chapter 13 — The five drawdown strategies

The **Drawdown** tab answers a complementary question: "*Given my portfolio, what do established withdrawal strategies say I can safely spend?*" Five deterministic strategies plus Monte Carlo are available.

- **4% Rule** — withdraw a fixed % of your starting portfolio in year 1, then inflation-adjust. Slider 2.5%–6%.
- **Variable Percentage Withdrawal (VPW)** — an increasing % calibrated to growth and horizon; reaches zero at life expectancy.
- **Three-Bucket** — cash funds years 1–3, Roth/low-risk funds years 4–10, growth funds year 11+.
- **Floor & Upside** — separates guaranteed income from portfolio-dependent income with a coverage table.
- **Guardrails** — starts at 5% with rules to cut/raise spending 10% as markets move.

Plain-language insights

Every strategy includes an Insights & Recommendations section in plain English — e.g. "*Even in the worst case, income would only drop to \$5,800/mo — within 15% of your current plan.*"



Chapter 14 — Stress Test: what 12 scenarios reveal

The **Stress test** tab runs your plan through 12 combined scenarios: four inflation rates (2%, your base, 5%, 8%) crossed with three return rates (bear -3%, your base, bull +3%). Each cell shows average monthly net income (nominal and real) and projected final-phase portfolio value.

Reading the grid

Your base case is highlighted. Upper-right cells (higher inflation/returns) tend to favor your plan; lower-left cells are the risk scenarios. If your plan looks healthy in the bottom-left corner, your strategy is robust to simultaneous adverse conditions.

Nominal vs real

Big nominal drops in high-inflation cells aren't necessarily bad — check the real number for true purchasing-power impact.

INFLATION & RETURNS STRESS TEST

▶ Stress test - 2 min

▶ Inflation - 2 min

Average net monthly income and estimated total portfolio at age 60 across 12 combined scenarios. Blue = your current base case. Return delta applies equally to 401k, cash, equity, Roth, and super.

	BEAR (-3%)	BASE RETURNS	BULL (+3%)
Low inflation (2%)	\$4,144/mo Real: \$3,519/mo Total at 60: \$704,569	\$4,138/mo Real: \$3,514/mo Total at 60: \$1,384,992	\$4,133/mo Real: \$3,510/mo Total at 60: \$2,599,269
Base inflation (3%)	\$4,145/mo Real: \$3,259/mo Total at 60: \$704,569	\$4,134/mo Real: \$3,250/mo Total at 60: \$1,384,992 ▲ BASE CASE	\$4,129/mo Real: \$3,246/mo Total at 60: \$2,599,269
High inflation (5%)	\$4,136/mo Real: \$2,809/mo Total at 60: \$704,569	\$4,130/mo Real: \$2,805/mo Total at 60: \$1,384,992	\$4,125/mo Real: \$2,802/mo Total at 60: \$2,599,269
Very high inflation (8%)	\$4,108/mo Real: \$2,286/mo Total at 60: \$704,569	\$4,102/mo Real: \$2,283/mo Total at 60: \$1,384,992	\$4,098/mo Real: \$2,280/mo Total at 60: \$2,599,269

Twelve scenarios at once: how your plan holds up across inflation and return combinations.

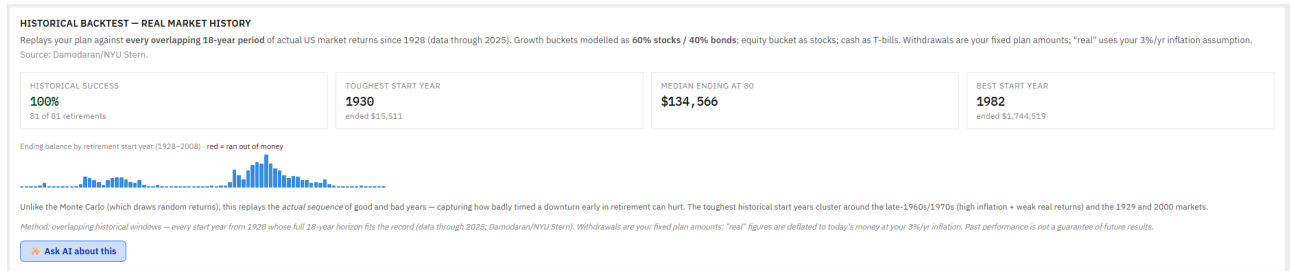
Historical backtest — would your plan have survived real history? NEW

Below the 12-scenario grid, the **Historical backtest** card asks a different question. Monte Carlo (Chapter 15) draws each phase's returns at random; the backtest replays the *actual sequence* of US market returns — every overlapping retirement window since **1928**. That sequence matters: a crash in your first few retirement years does far more damage than the same crash later, and random sampling tends to understate that "sequence-of-returns" risk. It runs automatically — no button to press.

Your **equity** bucket is modeled as 100% stocks, your **cash** bucket as Treasury bills, and your **growth** buckets (401k, Roth, Super) as a stock/bond blend you set with **Backtest growth-bucket stock %** (default 60/40; Chapter 7). Returns come from the Damodaran / NYU Stern public dataset, updated through the most recent complete year. Four tiles summarize the run: **historical success** (windows that paid every withdrawal, color-coded green ≥90% / amber 70–89% / red below), **toughest start year**, **median ending** balance, and **best start year**. A strip of bars below shows the ending balance for each start year, with red bars flagging windows that fell short. It replays returns only, not past inflation.

Backtest vs. Monte Carlo — use both

Monte Carlo asks "across hundreds of random futures, how often does my plan work?" The backtest asks "if the next decades look like any we've actually lived through, would I have made it?" A confident plan does well on both. If the backtest is noticeably weaker, sequence risk is your vulnerability — consider a larger early-years cash buffer or a slightly lower early withdrawal. (If your plan horizon is longer than the available history, the card says so rather than showing partial results.) An ✨ **Ask AI about this** button hands the result to the AI.



Replay your plan against every retirement since 1928 — the toughest start years show exactly where sequence risk bites.

Chapter 15 — Monte Carlo simulation

Monte Carlo runs up to 2,000 random-walk simulations (default 500). In each, a return is drawn for every account in each phase, and inflation for each year, from a normal distribution with mean equal to your configured rate and standard deviation equal to your configured sigma.

Reading the fan chart

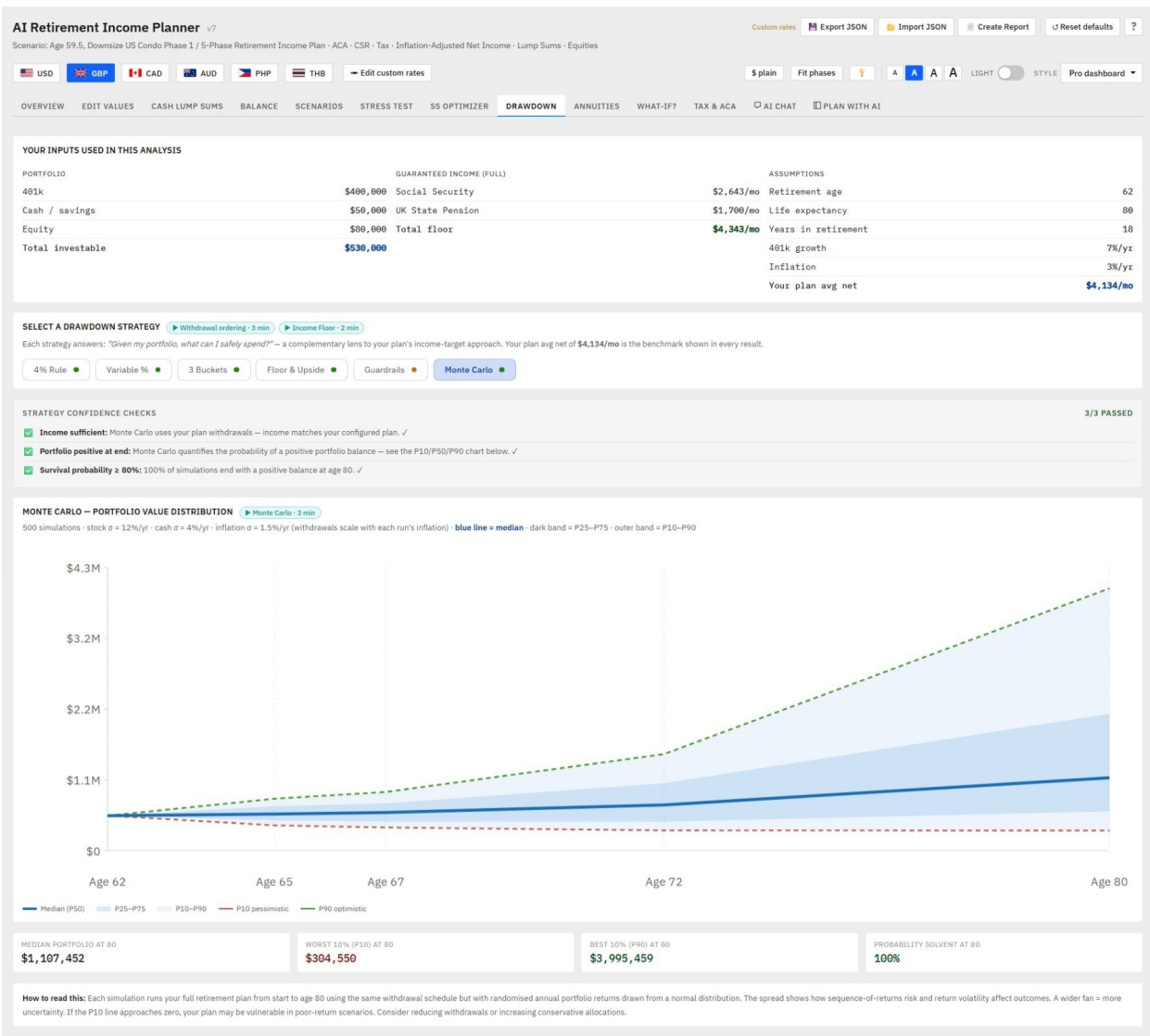
The chart shows percentile bands P10/P25/P50/P75/P90. The headline number is the probability of solvency — the share of runs in which your accounts pay every withdrawal to your final age.

Volatility settings

Each account has its own sigma on the Edit values tab: equity typically 12–18%, 401k/bonds 6–10%, cash 2–4%. The new ⓘ popover on the cash/bond volatility field explains sensible ranges (Chapter 28).

What Monte Carlo doesn't do

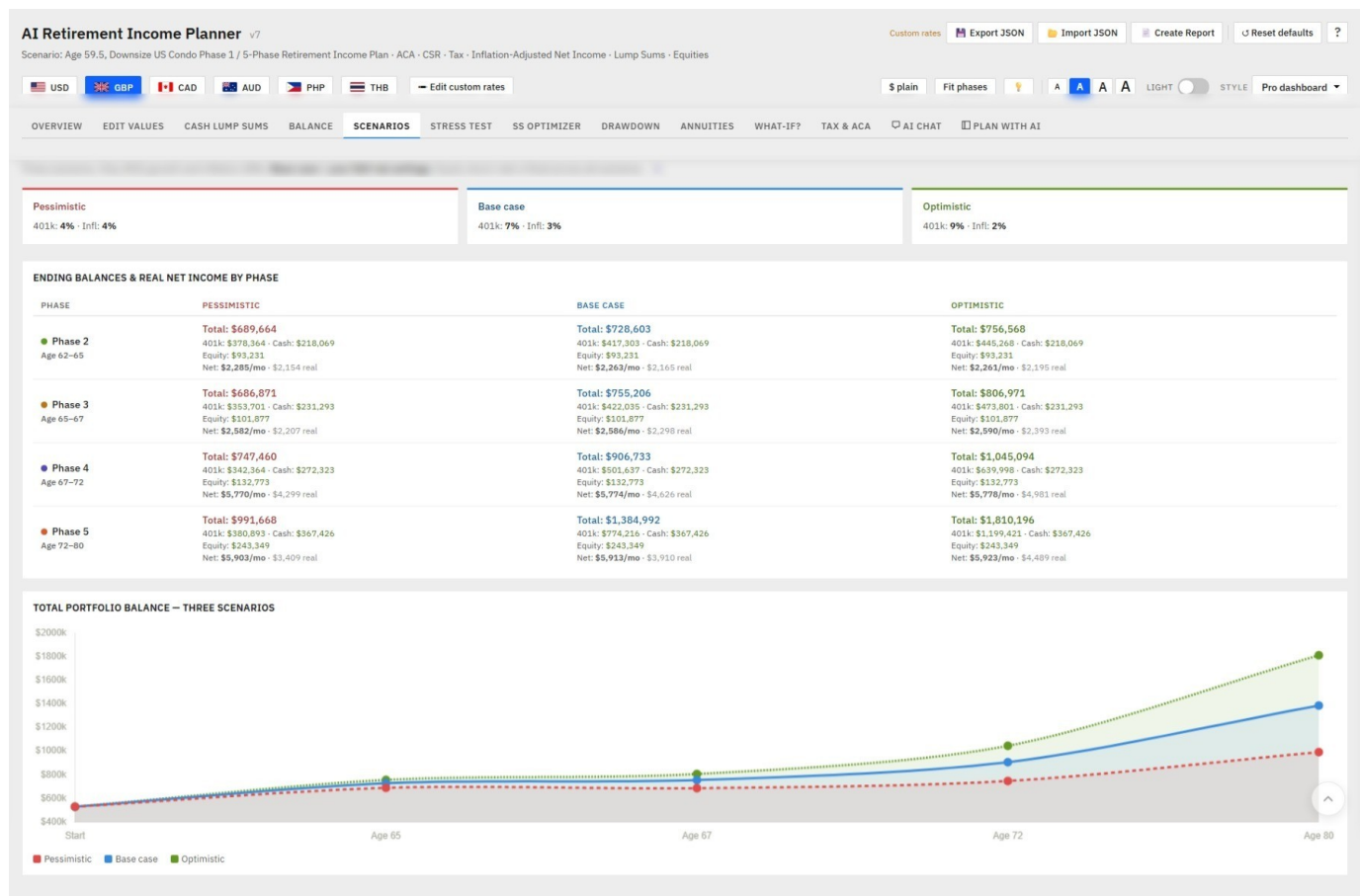
It does not predict the future — it quantifies uncertainty given your assumptions. A 92% probability of solvency means 92% of randomly-sampled future paths pay every withdrawal to the end.



The fan chart shows the range of outcomes; the headline figure is your probability of solvency.

Chapter 16 — Scenarios comparison

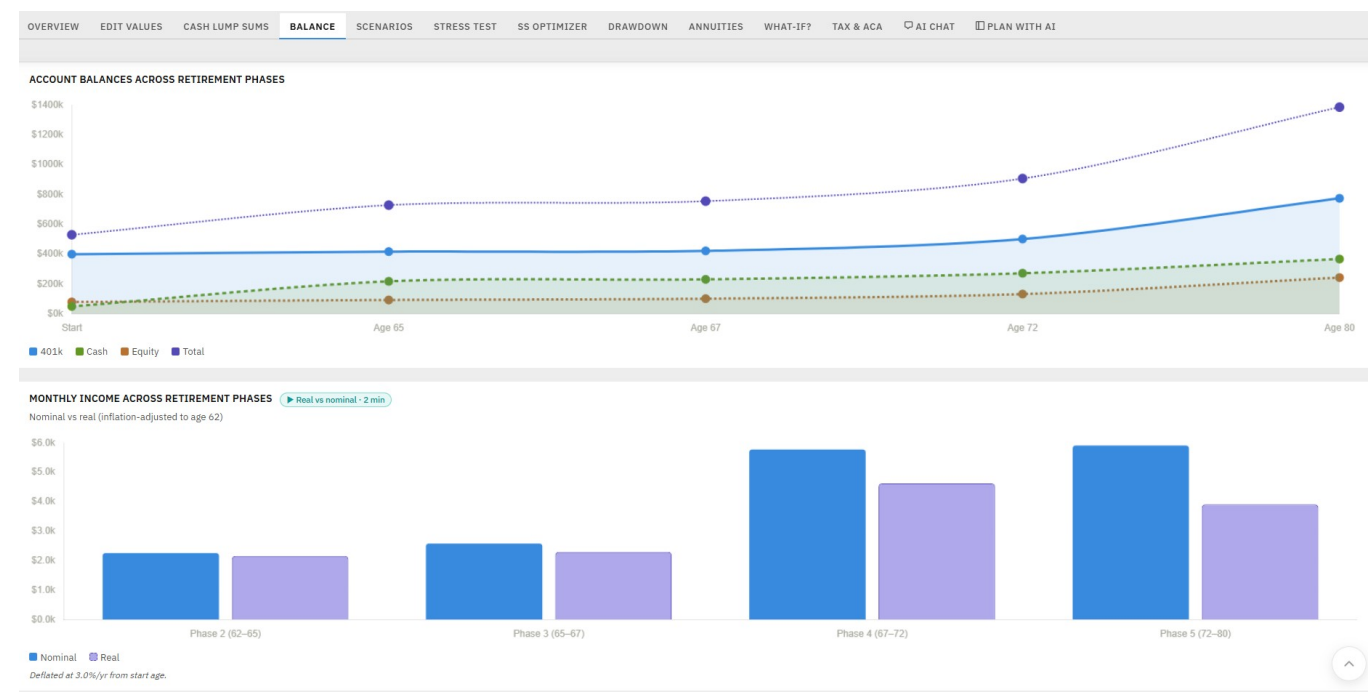
The **Scenarios** tab compares your plan against three pre-defined scenarios — Pessimistic (4% returns, 4% inflation), your Base case, and Optimistic (9% returns, 2% inflation) — side by side. Use it to see how much depends on your return assumption, how much real income you lose if inflation runs hot, and whether the pessimistic–optimistic gap is narrow enough to plan around.



Three return/inflation scenarios side by side.

Chapter 17 — Balance chart

The **Balance** tab visualizes your account balances (401k, Cash, Equity, Roth) over time, month by month. Each account is a colored line; the X axis is age. Look for lines crossing zero (accounts depleting early), steep drops, rising lines after retirement (growth outpacing withdrawal), and crossover points hinting at rebalancing opportunities.



Watch each account rise and fall across your whole retirement.

Part 4 — AI features

Several features use the AI provider you choose (Anthropic's Claude, OpenAI, Google Gemini, OpenRouter, or Ollama on your own computer) to make the planner more accurate, more conversational and more current. All are optional, and each needs a provider set up once (Chapter 18). In v7 the same setup also unlocks the new **Plan with AI** co-pilot (Chapter 20).

Chapter 18 — Choosing an AI provider and adding your key

The AI features work with the provider you choose: Anthropic (Claude), OpenAI, Google Gemini or OpenRouter, each with an API key from that company, or Ollama running on your own computer with no key at all. A key is different from a login: you create a developer account with the provider, generate a key, paste it into the planner once, and you're done. Cloud providers bill you for usage (typically a few cents per session); Ollama is free.

Where to get a key

- **Anthropic (Claude):** console.anthropic.com, under API Keys. The key starts with `sk-ant-`.
- **OpenAI:** platform.openai.com/api-keys. The key starts with `sk-`.
- **Google Gemini:** aistudio.google.com/apikey. The key starts with `AIza`.
- **OpenRouter:** openrouter.ai/keys. The key starts with `sk-or-`, and one key reaches models from many companies.
- **Ollama (local):** install it from ollama.com, download a model, and start Ollama with `OLLAMA_ORIGINS=*` set so the browser can reach it. No key is needed.

Cloud providers charge for what you use, so most accounts need a payment method or prepaid credit before a new key will answer. Check each provider's current pricing.

To set it up in the planner

1. Open the **AI Chat** tab (or the **Plan with AI** sidebar) and choose your **Provider**. For Ollama, check the **Endpoint URL** instead of entering a key.
2. Paste your **API key** and choose how the planner keeps it on this device: **Encrypt with a password** (your own computer), **Don't save it — this session only** (a shared or public one) or **Save without a password**.
3. Click **Save Key & Start Chatting**. Then pick a **Model** or keep the one suggested; ↻ **Refresh** reloads the provider's current list, and you can also type any model id.
4. Each provider keeps its own key and model, so you can switch between them without re-entering either. **Remove key** deletes a stored key.

Which provider to choose

For plan-editing proposals the planner recommends an Anthropic (Claude) key. Local Ollama models suit Learn the planner mode, and with Ollama nothing leaves your computer. Only Anthropic can search the web when you fetch current rates (Chapter 21); other providers answer from what their model already knows, so review each proposed figure before applying it.

The list also offers Custom (OpenAI-compatible), but the planner's security policy only allows the built-in providers' addresses, so most custom endpoints will not connect.

What is sent, and where

Your key goes straight to the provider you chose and never passes through our servers. When you use an

AI feature, your plan's figures (balances, ages, tax inputs) go to that provider under your own account; OpenRouter passes them on to the model you pick. With Ollama they stay on your computer.

One setup, every AI feature

Your provider and key power AI Chat, the Plan with AI sidebar, the state-tax lookup, Fetch current rates, the Help content audit, and AI commentary in the PDF report. Set it up once.

Keep your key private

Treat your API key like a password. Don't paste it into emails, chats or screenshots you share. If a key is ever exposed, delete it in your provider's account and create a new one.

To get started, choose your **AI provider** and enter your API key. Get a key at console.anthropic.com. [▶ API key setup](#)

Provider

Anthropic (Claude) ▼

API key

API key (sk-ant-...)

How should we keep your key on this device?

☒ 🔒 **Encrypt with a password** *(recommended for your own device)*
Saved scrambled; enter your password once per browser session to unlock AI.

☐ 👁 **Don't save it — this session only** *(recommended on a shared or public computer)*
Kept only until you close the browser; re-enter the key next time.

☐ 📄 **Save without a password** *(simplest; a private device you trust)*
Readable by anyone who uses this computer.

Choose a password (min 6 characters)

Forget it and you'll simply re-enter your API key — nothing is lost. The password protects the key on this device; it can't protect against a compromised browser.

Save Key & Start Chatting

Your key is stored only in your browser and sent directly to **Anthropic** — it never passes through any other server. When you use AI features, your plan's figures (balances, ages, tax inputs) are sent to Anthropic under your own account.

Choose a provider, paste its key, and decide how the planner keeps it. The key link and privacy note change with the provider you pick.

Chapter 19 — AI Chat: six starter prompts and how to read proposals

The **AI Chat** tab is the full-screen conversational surface. Every message includes a complete live snapshot of your plan — balances, phases, tax parameters, calculated results and Plan Health checks — so responses are specific to your numbers.

Six conversation-starter pills

- **Summarise & flag risks** — a 60-second overview with the 2–3 biggest risks.
- **Tax bracket check** — per-phase bracket distance and unused headroom.
- **ACA / CSR check** — cliffs and CSR opportunities in pre-Medicare phases.
- **Withdrawal strategy** — suggested adjustments to your account mix.
- **Propose optimizations** — specific numeric changes you can apply with one click.
- **Roth conversion advice** — conversion amounts that fit your bracket headroom.

AI proposals — applying changes directly

When you ask for optimizations, the AI returns a structured proposal (e.g. "*Reduce 401k withdrawal in Phase 2 by \$3,000, increase Roth by \$3,000 — keeps you below 400% FPL*") with an **Apply** button. AI proposals are constrained by your Plan Health state — the AI won't push you over the ACA cliff or into a higher IRMAA tier unless you explicitly ask.

New in v7 — copy any AI response

Each AI response now has a **copy** icon at the bottom. Click it to copy a clean, plain-text version of the reply to your clipboard — ready to paste into Word, email or your planning notes.

OVERVIEW
EDIT VALUES
CASH LUMP SUMS
BALANCE
SCENARIOS
STRESS TEST
SS OPTIMIZER
DRAWDOWN
ANNUITIES
WHAT-IF?
TAX & ACA
AI CHAT
PLAN WITH AI

AI RETIREMENT ASSISTANT
AI will receive your full plan on first message
Clear
Settings

Phase 2 (62-65): Modest lift in cash + equity draws to use the house-sale proceeds and raise real income while staying well under CSR ceiling (\$40,899) and far below the 12% bracket.

Phase 3 (65-67): Small lift – still under CSR (irrelevant post-65) and 12% bracket.

Phase 4 (67-72): Increase 401k to ~\$2,200/mo. Pre-empt RMD problem by drawing the 401k down before RMDs hit, and keeps you in 12% bracket with ~\$89k headroom. Lift cash draw to spend down the surplus.

Phase 5 (72-80): Raise 401k to \$1,600/mo to **clear the RMD requirement** (~\$1,526) with a small margin. Lift cash to draw down the surplus. Stays well within 12% bracket.

Expected Impact

- X RMD Compliance – ✓ (Phase 5 401k draw now exceeds RMD estimate)
- 12% bracket preserved in all phases (largest taxable income still = \$55k vs. \$62k+ ceiling)
- ACA CSR preserved in Phase 2 (MAGI stays under \$40,899)
- IRMAA preserved (largest MAGI = \$90k vs. \$136k Phase 4 threshold)
- Real income lifted meaningfully in every phase
- Portfolio still survives to 80 with margin (draws remain well within bucket budgets)

Health checks preserved: Income Adequacy, Tax Bracket Efficiency, ACA Subsidy Protected, IRMAA Not Triggered, Portfolio Survives to End, Income Floor Foundation, Stress Test Resilient

Checks improved: RMD Compliance (X – ✓ expected)

The app will now simulate these – If any check flips or RMD still falls short, I'll revise.

Ask about your retirement plan... (Shift+Enter for new line, Enter to send)
Send

AI Proposals
Review the AI's suggested changes. Uncheck any you don't want, then Apply.

☒
PHASE 2 CASH WITHDRAWAL /MO
\$150 → \$600
USE HOUSE-SALE LUMP SUM; TAX-FREE PRINCIPAL; KEEPS MAGI SAFELY UNDER ACA CSR CEILING \$40,899

☒
PHASE 2 EQUITY WITHDRAWAL /MO
\$350 → \$500
EQUITY BUCKET IS UNDERDRAWN AT 10% GROWTH; MOSTLY BASES, LOW TAX IMPACT, STAYS UNDER CSR

☒
PHASE 3 401K WITHDRAWAL /MO
\$2,200 → \$2,600
PRE-RMD BRACKET-FILLING AT 10-12%; REDUCES FUTURE RMD PRESSURE; STILL WELL UNDER 12% CEILING

☒
PHASE 3 CASH WITHDRAWAL /MO
\$150 → \$500
SPEND DOWN THE INFLATED CASH BUCKET; TAX-FREE; NO MAGI IMPACT

☒
PHASE 4 401K WITHDRAWAL /MO
\$1,350 → \$2,200
LOWERS PHASE 5 RMD BASE; USES ~\$89K HEADROOM TO 22% BRACKET; STAYS IN 12% BRACKET AND CLEAR OF IRMAA

☒
PHASE 4 CASH WITHDRAWAL /MO
\$100 → \$600
CONTINUE DRAWING DOWN CASH SURPLUS; TAX-FREE; NO BRACKET/IRMAA EFFECT

☒ SELECT ALL

Ask for optimizations, review the proposed changes, apply with one click — and copy the explanation to keep.

Chapter 20 — Plan with AI: the co-pilot sidebar NEW

The headline feature of v7 is **Plan with AI** — a dockable AI co-pilot that lives in a sidebar so you can keep your dashboard, phase cards or charts fully visible while you ask the AI to change your plan in plain English. Think of it as a planning assistant that sits beside your work, like the AI helpers in modern coding tools.

Opening and docking the sidebar

Click **Plan with AI** at the right-hand end of the tab row. The sidebar slides in from the left; your dashboard shifts over so nothing is hidden. The input box is always visible at the bottom. Click the button again (or the collapse control) to tuck it away. Your open/closed preference is remembered between sessions. On a phone the sidebar overlays the page and closes when you tap outside it.

A separate conversation from the AI Chat tab

The sidebar keeps its *own* conversation history, independent of the AI Chat tab — so a quick "bump SS to 67" in the sidebar won't clutter a longer analysis you're running in the tab. It has its own **clear-chat** button, a **copy** icon on responses, and an ⓘ **info** link to the in-app help for the sidebar.

Action-oriented quick chips

When the sidebar is empty it shows action chips that *change your plan* rather than just answer questions — for example "*rebalance so equity can grow*" or "*optimize my pre-Medicare phases for ACA*". The first, most prominent chip is a getting-started prompt: "*Start here: add your monthly withdrawals in Edit values or directly on the dashboard.*"

Verified proposals — the AI tests its own advice

This is what makes the co-pilot trustworthy. Because the planner's calculation engine runs instantly on your computer, the sidebar **tests every proposal in the simulator before you apply it**:

1. It applies the AI's proposed changes to a private copy of your plan and re-runs the full month-by-month projection.
2. It reads the result and shows a small **outcome badge** on the proposal — green if your plan stays solvent to your plan horizon, amber/red if any account would still run dry.
3. If you asked for a plan that doesn't run out of money and the first attempt still depletes, the co-pilot **automatically tells the AI its suggestion failed and asks it to refine** — up to three attempts — posting a short note each time it retries. You see only verified, workable proposals.

Why this matters

Language models can't reliably do month-by-month compound math in their heads. By checking each proposal against the planner's deterministic engine, the co-pilot won't hand you a confident-sounding suggestion that quietly runs you out of money in Phase 5.

Apply — and one-click Undo

Proposals appear as cards with every change pre-ticked; untick anything you don't want and click **Apply Selected**. The dashboard updates instantly. If you change your mind, a single **Undo** button reverses the entire batch you just applied — so you can experiment freely without manually putting numbers back.

Stays on topic

The co-pilot is focused on retirement planning. Ask it something unrelated and it will politely steer back to your plan.

AI Assistant

Quick questions to get started:

✓ Rebalance to grow equity

Optimise pre-Medicare for ACA

Fill the 12% bracket

Build a Roth conversion ladder

Optimise & set SS claim age

Smooth income across phases

Sidebar chat cleared. Use the quick questions below or ask anything.

Ask about your plan... (Enter to send)

AI Retirement Income Planner v7

Custom rates Export JSON Import JSON Create Report Reset defaults

Scenario: Age 59.5, Downsize US Condo Phase 1 / 5-Phase Retirement Income Plan · ACA · CSR · Tax · Inflation-Adjusted Net Income · Lump Sums · Equities

USD GBP CAD AUD PHP THB Edit custom rates

\$ plain Fit phases A A A LIGHT STYLE Pro dashboard

OVERVIEW EDIT VALUES CASH LUMP SUMS BALANCE SCENARIOS STRESS TEST SS OPTIMIZER DRAWDOWN ANNUITIES WHAT-IF? TAX & ACA AI CHAT PLAN WITH AI

AVG NET / MO \$4,134 nom - \$3,250 real EST. TOTAL AGE 80 (USD) \$1,384,992 PHASE 5 PROJECTION: Age 72-80 Extend to age 90 Total tax to 88: \$61,879 Plan Health: 7/8 Replan from today

REPLAN FROM CURRENT AGE Replan · 90 sec

First time using Replan? Enter today's reality — your actual current age, current account balances, and current SS / UKP payment amounts (the amounts you receive now, after any COLA / triple-lock growth that has already happened). Phases you've already lived through will be trimmed and the rest of the plan re-anchored from your real situation. Your original plan stays saved until you Apply — and you can roll back any time with Reset to original plan.

CURRENT AGE e.g. 63

ACTUAL 401K BALANCE (USD) 480000

ACTUAL CASH BALANCE (USD) 50000

ACTUAL EQUITY BALANCE (USD) 80000

✓ Apply & recalculate

Cancel

Reset to original plan

STARTING 401K (USD) \$400,000 STARTING CASH (USD) \$50,000 STARTING EQUITY (USD) \$80,000

RETIREMENT PHASES

PHASE 2 Age 62-65 · 36mo

401k withdrawal \$1,980/mo

Cash withdrawal \$150/mo

Equity withdrawal \$350/mo

House Sale \$150,000 one-time

Gross income \$2,400/mo

Federal income tax -\$58/mo

ACA premium est. -\$79/mo

✓ ACA subsidy eligible

MAGI > 100% FPL

100-250% FPL

PHASE 3 Age 65-67 · 24mo

401k withdrawal \$2,250/mo

Cash withdrawal \$150/mo

Equity withdrawal \$450/mo

Part-time work \$1,200/yr

Gross income \$2,950/mo

Federal income tax -\$83/mo

Medicare Part B est. -\$228/mo

Medicare Part D est. -\$53/mo

10% bracket (\$13,956) -2030 est.

PHASE 4 Age 67-72 · 60mo

401k withdrawal \$1,350/mo

Cash withdrawal \$100/mo

Equity withdrawal \$450/mo

US Social Security \$2,784/mo

UK State Pension \$1,085/mo

Gross income \$6,489/mo

Federal income tax -\$403/mo

Medicare Part B est. -\$253/mo

Medicare Part D est. -\$50/mo

PHASE 5 Age 72-80 · 96mo

401k withdrawal \$800/mo

Cash withdrawal \$0 /mo

Equity withdrawal \$350/mo

US Social Security \$3,293/mo

UK State Pension \$2,191/mo

Gross income \$6,634/mo

Federal income tax -\$342/mo

Medicare Part B est. -\$307/mo

Medicare Part D est. -\$71/mo

The Plan with AI co-pilot docks on the left so your dashboard stays in view.

AI Assistant

BOOST EARLY REAL INCOME

PHASE 3 401K /MO

\$2,250 → \$2,800

PRE-SS BRIDGE; AMPLE IRMAA HEADROOM

PHASE 3 CASH /MO

\$150 → \$800

CONTINUE CASH SPEND-DOWN BEFORE SS ARRIVES

PHASE 3 EQUITY /MO

\$450 → \$600

SMOOTH INCOME INTO MEDICARE TRANSITION

PHASE 4 401K /MO

\$1,350 → \$900

SS+UKP COVER MOST OF NEED; REDUCE PORTFOLIO DRAW

PHASE 4 EQUITY /MO

\$450 → \$250

CUT TO DAMPEN POST-SS INCOME SPIKE

PHASE 5 401K /MO

\$800 → \$1,550

MEET RMD REQUIREMENT (~\$1,526/MO) AND LEVEL REAL INCOME

PHASE 5 EQUITY /MO

\$350 → \$150

REDUCE TO MATCH SMOOTHED LIFESTYLE LEVEL

✓ Verified by simulation — no accounts deplete after these changes.

Apply Selected

Dismiss

Ask about your plan... (Enter to send)

➔

AI Retirement Income Planner v7

Custom ratesExport JSONImport JSONCreate ReportReset defaults?

Scenario: Age 59.5, Downsize US Condo Phase 1 / 5-Phase Retirement Income Plan - ACA - CSR - Tax - Inflation-Adjusted Net Income - Lump Sums - Equities

USDGBPCADAUDPHPTHB

Edit custom rates

\$ plainFit phasesLIGHTSTYLEPro dashboard

OVERVIEWEDIT VALUESCASH LUMP SUMSBALANCESCENARIOSSTRESS TESTSS OPTIMIZERDRAWDOWNANNUITIESWHAT-IF?TAX & ACATAI CHATPLAN WITH AI

AVG NET / MO

\$4,134nom→\$3,250real

EST. TOTAL AGE 80 (USD)

\$1,384,992

PHASE 5 PROJECTION: Age 72-80Extend to age 90

Total tax to 80: \$61,879

Plan Health: 7/8✓

Replan from today

REPLAN FROM CURRENT AGE

Replan - 90 sec

First time using Replan? Enter today's reality — your actual current age, current account balances, and current SS / UKP payment amounts (the amounts you receive now, after any COLA / triple-lock growth that has already happened). Phases you've already lived through will be trimmed and the rest of the plan re-anchored from your real situation. Your original plan stays saved until you Apply — and you can roll back any time with Reset to original plan.

CURRENT AGE

e.g. 63

ACTUAL 401K BALANCE (USD)

400000

ACTUAL CASH BALANCE (USD)

50000

ACTUAL EQUITY BALANCE (USD)

80000

✓ Apply & recalculate

Cancel

Reset to original plan

STARTING 401K (USD)

\$400,000

STARTING CASH (USD)

\$50,000

STARTING EQUITY (USD)

\$80,000

RETIREMENT PHASES

PHASE 2Age 62-65 · 36mo

401k withdrawal\$1,900/mo

Cash withdrawal\$150/mo

Equity withdrawal\$350/mo

House Sale\$150,000 one-time

Gross income\$2,400/mo

Federal income tax-\$58/mo

ACA premium est.-\$79/mo

✓ ACA subsidy eligibleMAGI > 100% FPL

ACA Silver PDB100-240% FPL

PHASE 3Age 65-67 · 24mo

401k withdrawal\$2,250/mo

Cash withdrawal\$150/mo

Equity withdrawal\$450/mo

Part-time work\$1,200/yr

Gross income\$2,950/mo

Federal income tax-\$83/mo

Medicare Part B est.-\$228/mo

Medicare Part D est.-\$53/mo

10% bracket (\$13,956)~2030 est.

PHASE 4Age 67-72 · 60mo

401k withdrawal\$1,350/mo

Cash withdrawal\$100/mo

Equity withdrawal\$450/mo

US Social Security\$2,784/mo

UK State Pension\$1,885/mo

Gross income\$6,489/mo

Federal income tax-\$403/mo

Medicare Part B est.-\$253/mo

Medicare Part D est.-\$50/mo

PHASE 5Age 72-80 · 96mo

401k withdrawal\$800/mo

Cash withdrawal\$0 /mo

Equity withdrawal\$350/mo

US Social Security\$3,293/mo

UK State Pension\$2,191/mo

Gross income\$6,634/mo

Federal income tax-\$342/mo

Medicare Part B est.-\$307/mo

Medicare Part D est.-\$71/mo

Each proposal is tested in the simulator first — green means your plan still lasts to your plan horizon. Undo reverses the whole batch.

Chapter 21 — Fetch current rates (and state-tax lookup)

Tax brackets, FPL thresholds, the IRMAA ladder, Medicare premiums and the standard deduction all change every year, and a file you downloaded months ago still holds the figures it shipped with. The **Fetch current rates** button at the bottom of the Edit values tab handles them in one click.

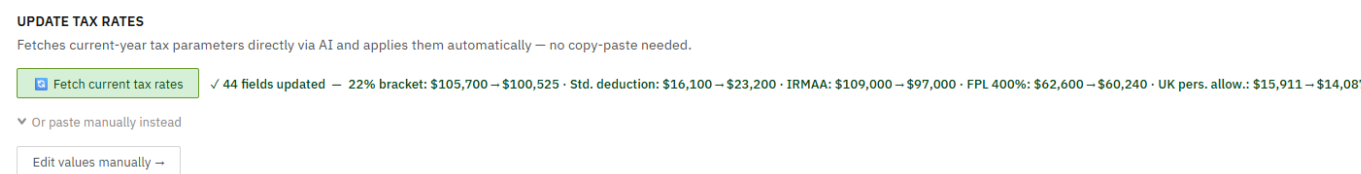
1. On the Edit values tab, scroll to the bottom and find **Fetch current rates**.
2. Click it. The AI looks up current-year values for federal brackets, the poverty levels behind ACA subsidies, the full six-bracket IRMAA ladder, the standard and senior deductions, Medicare Part B and Part D premiums, the Social Security earnings-test limits, and UK / Canadian / Australian parameters. Where your AI provider supports it the lookup searches the web, which matters because every one of these figures is reset annually and a model answering from memory will give you last year's. It asks only for the figures your plan can use unless you tick Ask for every country.
3. A review table appears listing every figure it wants to change, what you have now beside what it proposes, and a tick box against each one. Anything that fails a plausibility check — a value outside the range that figure can sensibly take, brackets that do not rise in order, a jump far larger than a year's indexation — arrives already unticked with the reason beside it.
4. Press **Apply** and only the ticked rows are written — nothing changes until you do.

Without an API key

The same area shows the **Rate Research Prompt** — a paste-ready prompt for any external AI (Claude.ai, ChatGPT, Gemini). Copy it, paste in any AI, copy the response back, and the planner imports the values.

New in v7 — state income-tax lookup

Alongside the federal fetch, the **State** field on the Tax parameters card has its own **Auto-fill rate** button that looks up your state's flat-rate approximation and retirement-income exemptions (see Chapter 7). It's enabled only once you've entered a state.



One click proposes every tax parameter, shows you exactly what would change, and waits for you to accept it.

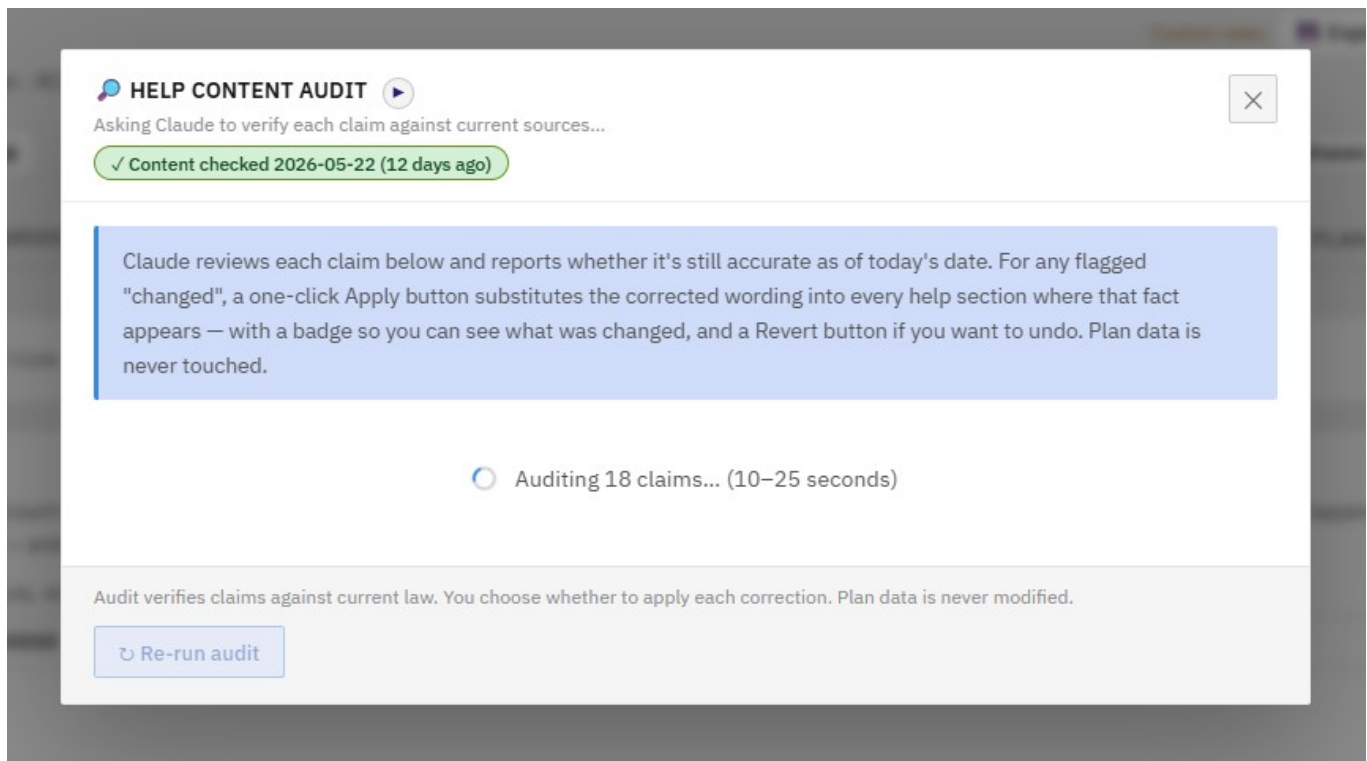
Chapter 22 — Help content audit and one-click corrections

Tax brackets are one form of drift; structural rules and dates (RMD age, ACA cliff status, IRMAA mechanics, treaty articles, recent COLA figures) are another. The **Help content audit** handles those — and can apply corrections to the planner's help prose with a single click.

To run an audit

1. Click **Audit help content** in the planner header.
2. Wait 10–25 seconds while the AI reviews 15 specific factual claims against today's law.
3. Each claim is shown as *current*, *changed*, or *uncertain*.
4. For any *changed* row, click **Apply this update everywhere** — the corrected prose is substituted into the Tax & ACA notes, Concepts Primer, info popovers and glossary, each with a small *AI-corrected* badge.
5. Click **Revert** on any row to restore the original prose.

If all claims come back current, you'll see a green banner and a persistent header badge showing the last audit date. After ~18 months the badge turns amber as a gentle stale reminder. Corrections survive reload and travel with your plan in JSON.



The AI fact-checks 15 claims against today's law — apply corrections everywhere with one click.

Chapter 23 — The PDF report (now with allocation, resilience & charts)

The **Report** button produces a clean, multi-page reference report you can keep, email or hand to an advisor. With an API key you can opt in to AI-enhanced commentary that adds short interpretive prose without inventing any numbers. Version 7 substantially expands what the report contains.

What's in the report

- **Cover** — scenario title, generation date, currency, residency, headline metrics, and (optional) a 60-word AI plan summary.
- **Phase-by-phase plan** — one printable card per phase with income sources, tax, healthcare and balances.
- **Top risks & recommended actions** — drawn from your Plan Health failures, with optional AI narratives.

Asset-allocation mix v7 — your portfolio composition at the start and end of the plan, both by account (401k / Cash / Equity / Roth / Super) and by tax treatment (tax-deferred / taxable / tax-free), with a one-line tax-diversification note.

Plan-resilience page v7 — opens with a plain-English **Confidence summary** that combines all three lenses (Plan Health checks passed, the Monte Carlo success rate, and the historical backtest result) and names the main thing to watch — followed by the Monte Carlo detail ("succeeded in N% of simulations to age X" with P10 / median / P90 ending portfolios), the 12-scenario stress-test grid with your base case highlighted, and a historical-backtest summary (how many real retirements since 1928 the plan survived, with the toughest start year).

Portfolio charts v7 — the balance and income charts rendered into the report.

- **Key assumptions** and a **disclaimer** (noting that Monte Carlo / stress results are scenario analyses, not guarantees). Three further sections are optional, each a tick in the report toolbar and all three off until you set them. Include notes prints the plan notes you wrote in the Saved plans panel as their own page, and stays disabled until the plan has notes in it. Include calculations adds the working behind every phase — how the tax thresholds were inflated to that phase, federal tax bracket by bracket, the Social Security provisional-income test, how Medicare was estimated, and the line from gross income to what you keep — at roughly three pages per phase, so a five-phase plan about doubles in length. Include AI commentary adds the interpretive prose described below, and stays disabled until an API key is saved in the AI Chat tab.

Fixed in v7 — no more blank pages

Earlier versions could emit blank or clipped pages when printing the report. v7 rebuilds the report to flow

naturally across pages, so the printed PDF matches the preview.

What the AI adds — and doesn't

AI commentary is interpretive prose only. It never generates new dollar figures (all numbers come deterministically from your plan), and the entire AI text appears in an inline-editable preview before going to PDF. Typical cost: about \$0.10 per export.

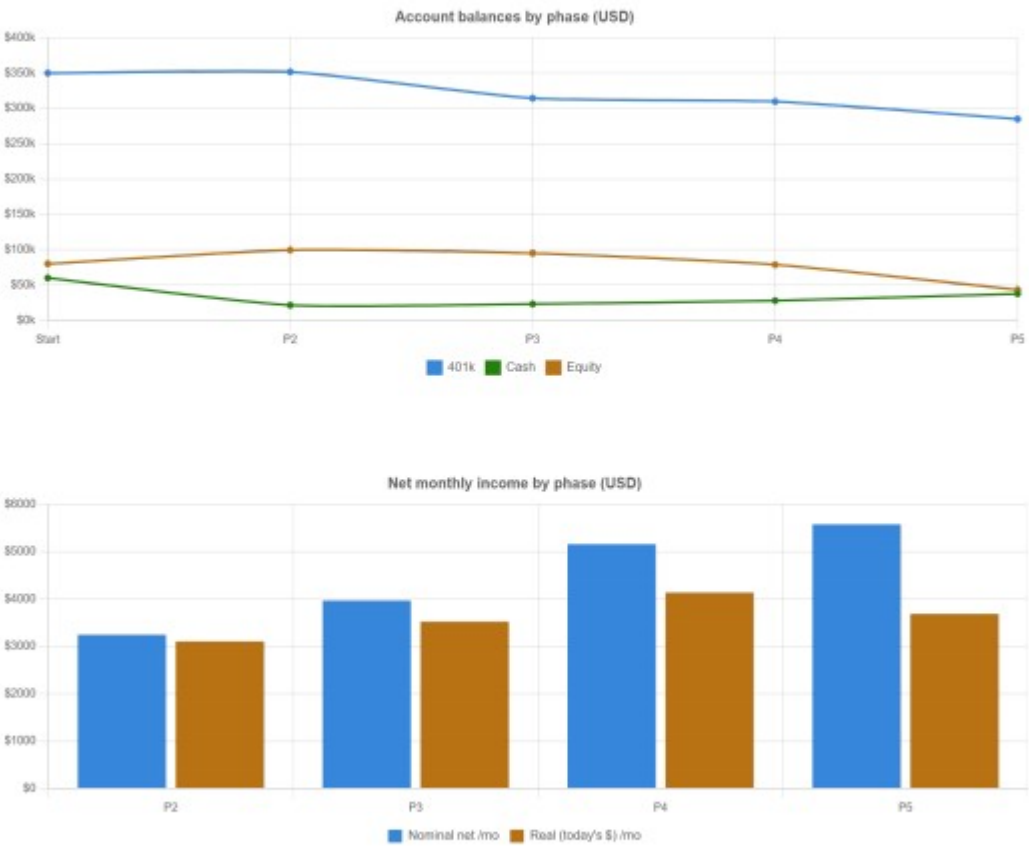
Balance trajectory

Projected account balances at the end of each phase, based on your withdrawal strategy and growth assumptions. A zero balance in an income-producing phase is flagged — see notes below.

PHASE	401K	CASH	EQUITY	TOTAL
Starting (age 62)	\$350,000	\$60,000	\$80,000	\$490,000
Phase 2 (ages 62–65)	\$351,664	\$21,484	\$99,498	\$472,646
Phase 3 (ages 65–67)	\$314,461	\$23,155	\$94,980	\$432,596

Portfolio charts

Visual summary of how your account balances and net income evolve across the plan.



Plan resilience

How the plan holds up under uncertainty — random return sequences (Monte Carlo) and combined inflation / return stress scenarios. These are scenario analyses, not guarantees.

Monte Carlo simulation

Across **500** randomized simulations to age 80, your plan ended solvent (money still remaining) in **100%** of runs — a strong result.

PROBABILITY SOLVENT

100%

WORST 10% (P10) AT 80

\$39,710

MEDIAN (P50) AT 80

\$394,292

BEST 10% (P90) AT 80

\$2,130,581

Each simulation draws annual returns for each account from a normal distribution around your assumed rates. Because the simulation is random, this percentage may vary by a point or two between reports.

v7's report adds allocation, resilience and charts — the kind of pages a professional advisor report includes.

Part 5 — Specialized tools

Several tabs go deeper than the core planner: the Annuity Calculator, the What-if Explorer, the SS Optimizer, and the Roth Conversion Optimizer integration.

Chapter 24 — Annuity Calculator

The **Annuities** tab models fixed annuity purchases — SPIA (immediate income), DIA / Longevity (deferred income), or Fixed + COLA (income that grows annually) — and compares them against keeping the money invested. Variable annuities are not modeled.

Setting up an annuity

1. Click the **Annuities** tab.
2. Choose annuity type: SPIA, DIA / Longevity, or Fixed + COLA.
3. Enter premium, purchase age, payout rate (or monthly income — both update together).
4. For DIA set the income-start age (typically 80 or 85); for Fixed + COLA set the COLA rider rate.
5. Choose coverage: Single life or Joint life.
6. Review monthly income, payout rate, break-even age, and the year-by-year comparison versus investing.

Adding to (and removing from) your plan

The **Add to Plan** section deducts the premium from your funding source(s) — cash only, or a custom split across cash, 401k, equity, Roth and any lump sum — and adds annuity income to passive income for each phase it's active in. Every lump sum in your plan appears as a fundable source; allocate part of a lump and the annuity sizes down to match. A **Remove annuity from plan** button reverses every change cleanly.

AI Retirement Income Planner v7

Custom ratesExport JSONImport JSONCreate ReportReset defaults?

Scenario: Age 59.5, Downsize US Condo Phase 1 / 5-Phase Retirement Income Plan · ACA · CSR · Tax · Inflation-Adjusted Net Income · Lump Sums · Equities

USDGBPCADAUDPHPTHBEdit custom rates

\$ plainFit phasesLightSTYLEPro dashboard

OVERVIEWEDIT VALUESCASH LUMP SUMSBALANCESCENARIOSSTRESS TESTSS OPTIMIZERDRAWDOWNANNUITIESWHAT-IF?TAX & ACATAI CHATPLAN WITH AI

ANNUITY CALCULATORIncome Floor · 2 min

Model a fixed annuity purchase — SPIA (immediate income), DIA / Longevity (deferred income), or Fixed + COLA. Compare against keeping the money invested, find the break-even age, and add to your plan when ready. Variable annuities are not modelled.

ANNUITY DETAILS

TYPE

SPIADIA / LongevityFixed + COLA

PREMIUM (USD)

200000

YOUR AGE AT PURCHASE

62

PAYOUT RATE (%/YR) OR MONTHLY INCOME

6or1000

Enter payout rate OR monthly income — both update together

COVERAGE

Single lifeJoint life

INVESTMENT RETURN FOR COMPARISON (%/YR)

6

RESULTS

MONTHLY INCOME

\$1,000

6.0% payout rate

BREAK-EVEN AGE

78

16 yrs from income start

ANNUAL INCOME

\$12k

COVERAGE

Single

Ends at death

CUMULATIVE INCOME RECEIVED

Age 75

\$168k

Not yet

Age 80

\$228k

✓ Ahead

Age 85

\$288k

✓ Ahead

Age 90

\$348k

✓ Ahead

Age 95

\$408k

✓ Ahead

Type: SPIA — income begins immediately · Coverage: Single life — income ends at death

ANNUITY VS INVESTING THE PREMIUM

Cumulative annuity income received vs what the \$200,000 premium would be worth if invested at 6%/yr. The annuity "wins" once cumulative income overtakes the invested value.

Age	Cumul. annuity income	Premium if invested at 6%	Status
62 (purchase)	\$12k	\$200k	—
65	\$48k	\$238k	—
70	\$108k	\$319k	—
75	\$168k	\$427k	—
80	\$228k	\$571k	✓ Ahead
85	\$288k	\$764k	✓ Ahead
90	\$348k	\$1.0M	✓ Ahead
95	\$408k	\$1.4M	✓ Ahead
100	\$468k	\$1.8M	✓ Ahead

ADD TO PLAN

Choose which account(s) the premium comes from, then review the impact. The annuity income gets added to Passive Income for each affected phase.

Funding source

Cash onlyCustom split

Premium will be deducted from cash. Cash available: \$50,000 · premium exceeds available cash by \$150,000

Changes preview

– \$200,000 from cash balance (at age 62)

+ \$12,000/yr passive income in Phase 2 (ages 62–65)

+ \$12,000/yr passive income in Phase 3 (ages 65–67)

+ \$12,000/yr passive income in Phase 4 (ages 67–72)

+ \$12,000/yr passive income in Phase 5 (ages 72–80)

✓ Add annuity to plan

Passive income fields live on the Edits tab. You can fine-tune amounts there after adding.

Model SPIA, DIA or COLA annuities and see the break-even age versus staying invested.

Chapter 25 — What-if Explorer (and the survivor scenario)

The **What-if?** tab is a sandbox for stress-testing specific decisions without modifying your actual plan. Several live tools let you explore the most common "what should I do?" questions — three sliders plus the new Maximum-sustainable-spending finder (below):

- **Social Security claim age (62→70)** — monthly benefit, cumulative payments by 85 and 90, and break-even age.
- **Roth conversion size (per phase)** — tax cost, MAGI shift, ACA / IRMAA status change, net income impact.
- **Inflation rate** — how a 1% change shifts real income at the end of your plan.

Move a slider and the right-hand panel updates instantly — no Calculate button, no waiting. Nothing here writes back to your saved plan.

New in v7 — survivor-scenario setup guidance

The survivor scenario (what happens when one spouse outlives the other) lives under Edit values → Income, where Who dies first decides whose pensions stop or shrink and whose age then sets Medicare, required withdrawals and the age-65 deduction. Its tax results are accurate once your filing status and US tax obligation are set; its popup's Take me there → link jumps there.

The screenshot shows the 'WHAT-IF EXPLORER' interface. At the top, it says 'Live calculators using your current plan numbers as the starting point. Pick a tool below to explore one decision at a time — nothing here changes your saved plan.' Below this are five buttons: 'Social Security', 'Roth conversion', 'Inflation', 'Survivor' (which is highlighted in blue), and 'Max spending'.

Under the 'Survivor' button, the section is titled 'SURVIVOR — WHEN ONE SPOUSE DIES'. The text reads: 'On the first death the household loses the smaller Social Security check and the survivor usually moves from Married-Filing-Jointly to **single** tax brackets — often a sharp drop in net income for barely-lower expenses (the "widow's tax cliff").' There is an information icon (i) next to this text. Below this, a smaller line of text says: 'Survivor analysis applies to Married-Filing-Jointly plans with both Social Security amounts entered.'

A popup window is open over the bottom right of the interface. It has a title bar 'Survivor scenario (when one spouse dies)' with an information icon (i) and a close button (X). The main text in the popup says: 'On the first death the household loses the **smaller Social Security check** and the survivor usually moves to **single** tax brackets — a sharp income drop for barely-lower expenses (the "widow's tax cliff").' Below this text is a yellow box with the following text: 'ⓘ **Set-up needed.** The survivor projection only runs for Married-Filing-Jointly plans with both Social Security amounts entered. In **Edit values → US tax obligation**, click **Married filing jointly**, then fill in your US Social Security and your spouse's SS (amount, claim age, and current age if there's an age gap).' At the bottom of the yellow box is a blue button that says 'Take me there →'. Below the yellow box, there is a link that says 'Read more in Glossary →'.

The survivor scenario now guides you to set up your US tax obligation first — with a one-click jump to that section.

Maximum sustainable spending NEW

This card answers the question behind every retirement plan — "*how much can I safely spend?*" It searches for the highest **withdrawal level** your plan can support by scaling every phase's withdrawals up or down together and testing each level in the real simulator. Your guaranteed income (Social Security, pensions) is left untouched, and — like the rest of this tab — it never changes your saved plan. Pick what "safe" means with the **Test against** selector: *Pays every withdrawal to plan end* (the default, instant), *Monte Carlo success*, or *Survives market history*. For the last two you can set the success **target %** (defaults: 85% for Monte Carlo, 90% for history) and click **Calculate**. The result shows your maximum sustainable net income versus your current plan average — so you can see at a glance whether you have room to spend more, or by how much to trim. An ✨ **Ask AI about this** button hands the figure to the AI for next steps.

It's an estimate, not a solver

The tool scales all withdrawals proportionally and treats the pass/fail point as a single boundary — a fast, transparent planning estimate, not a multi-variable optimizer. Use it to size the ballpark, then fine-tune individual phases on the Edit values tab.

 **WHAT-IF EXPLORER**

Live calculators using your current plan numbers as the starting point. **Pick a tool below** to explore one decision at a time — nothing here changes your saved plan.

Social Security

Roth conversion

Inflation

Survivor

Max spending

MAXIMUM SUSTAINABLE SPENDING

Answers "how much can I safely spend?" — it searches for the highest **withdrawal level** your plan can support, by scaling every phase's withdrawals up or down together and testing the result in the simulator. Your guaranteed income (Social Security, pensions) is untouched; this never changes your saved plan.

TEST AGAINST:

Stays solvent to plan end ▼

MAXIMUM SUSTAINABLE NET INCOME

\$5,741/mo vs \$5,408/mo in your current plan

≈ 113% of your current planned withdrawals · real (today's money) ≈ \$4,661/mo

You could raise withdrawals to about **113%** of today's level and the plan still keeps your drawn-from accounts positive to plan end — headroom of roughly **\$332/mo**.

 **Ask AI about this**

Method: a bisection search on a single withdrawal-level multiplier, tested in the real simulator (target: keeps your drawn-from accounts positive to plan end). It scales all phase withdrawals together and assumes spending changes are roughly proportional — a planning estimate, not a guarantee. Your saved plan is unchanged.

How much can you safely spend? A transparent search for your sustainable withdrawal level — solvent, Monte Carlo, or against real history.

Chapter 26 — Social Security Optimizer

The **SS Optimizer** tab evaluates starting Social Security at each eligible age from 62 to 70. For each starting age it shows cumulative lifetime benefits at three longevity targets — 80, 85 and 90 — and highlights the strategy that produces the most lifetime income at each.

Key insights

- Most people who reach 65 will live past 85 — which favors claiming later.
- Claiming at 62 only "wins" if you live to roughly the early 80s or sooner.
- Delaying to 70 maximizes monthly income for life but loses ~8 years of payments; break-even is typically late 70s to early 80s.
- Health, family longevity, and whether you need the money before 70 matter more than the spreadsheet number.

AI Retirement Income Planner v7

Custom ratesExport J50NImport J50NCreate ReportReset defaults?

Scenario: Age 59.5, Downsize US Condo Phase 1 / 5-Phase Retirement Income Plan - ACA - CSR - Tax - Inflation-Adjusted Net Income - Lump Sums - Equities

USDGBPCCADAUDPHPTHBEdit custom rates\$ plainFit phasesA A A LIGHTSTYLEPro dashboard

OVERVIEWEDIT VALUESCASH LUMP SUMSBALANCESCENARIOSSTRESS TESTSS OPTIMIZERDRAWDOWNANNUITIESWHAT-IF?TAX & ACACAI CHATPLAN WITH AI

Your Edit values tab has claiming age 67 with monthly benefit \$2,643/mo. Amounts at other claim ages are projected using SSA's actuarial reduction and delayed credit rules.

MONTHLY BENEFIT BY CLAIMING AGE

Based on your entered benefit of \$2,643/mo at age 67. Your Full Retirement Age (FRA): 67 (estimated — set your birth year on the Edit tab for precision). All figures in today's dollars (pre-COLA).

CLAIM AGE	PRIMARY /MO	CUMUL. TO 70	CUMUL. TO 75	CUMUL. TO 80	CUMUL. TO 85	CUMUL. TO 90	VS AGE 62
62	\$1,850/mo	\$177,610	\$288,616	\$399,622	\$510,628	\$621,634	—
63	\$1,982/mo	\$166,509	\$285,444	\$404,379	\$523,314	\$642,249	+\$132/mo
64	\$2,114/mo	\$152,237	\$279,101	\$405,965	\$532,829	\$659,693	+\$264/mo
65	\$2,291/mo	\$137,436	\$274,872	\$412,308	\$549,744	\$687,180	+\$441/mo
66	\$2,467/mo	\$118,406	\$266,414	\$414,422	\$562,430	\$710,438	+\$617/mo
67	\$2,643/mo	\$95,148	\$253,728	\$412,308	\$570,888	\$729,468	+\$793/mo
68	\$2,854/mo	\$68,507	\$239,773	\$411,039	\$582,306	\$753,572	+\$1,004/mo
69	\$3,066/mo	\$36,791	\$220,743	\$404,696	\$588,649	\$772,602	+\$1,216/mo
70	\$3,277/mo	—	\$196,639	\$393,278	\$589,918	\$786,557	+\$1,427/mo

BREAKEVEN AGES — DELAY VS. CLAIM AT 62

The age at which delaying SS pays off vs. claiming at 62. Does not account for investment returns on early benefits or COLA. A healthy individual with longevity in the family often benefits from delaying to 67 or 70.

CLAIM AT	VS. CLAIM AT 62	BREAKEVEN AGE	STRATEGY NOTE
63	+\$132/mo	Age 77.0	Good (breakeven 75–80)
64	+\$264/mo	Age 78.0	Good (breakeven 75–80)
65	+\$441/mo	Age 77.7	Good (breakeven 75–80)
66	+\$617/mo	Age 78.0	Good (breakeven 75–80)
67	+\$793/mo	Age 78.7	Good (breakeven 75–80)
68	+\$1,004/mo	Age 79.1	Good (breakeven 75–80)
69	+\$1,216/mo	Age 79.7	Good (breakeven 75–80)
70	+\$1,427/mo	Age 80.4	Moderate (breakeven 80–85)

Tip: If you plan to use Roth or cash withdrawals to bridge income from retirement to SS claiming age, delaying SS to 67 or 70 is often the highest guaranteed return available. Enter your chosen SS claiming age's benefit amount in the main Income streams section and adjust Phase 1/2 withdrawals to compensate for the lower early income.

Cumulative benefits at every claiming age, scored against three longevity targets.

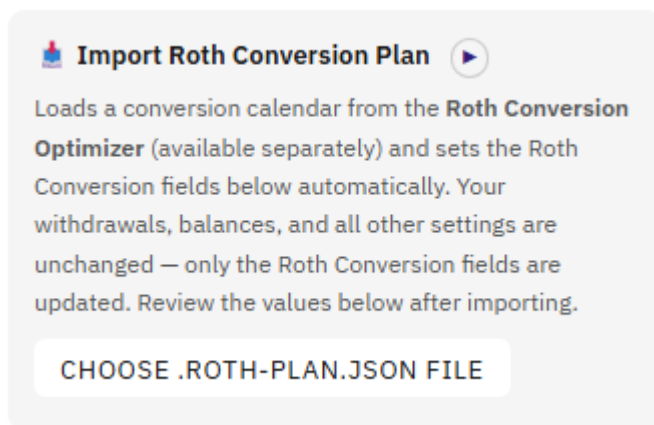
Chapter 27 — Roth Conversion Optimizer integration

Roth conversions are one of the highest-leverage tax decisions in retirement. The planner has its own per-phase Roth conversion field, but designing a multi-year schedule that fills bracket headroom while staying below ACA / IRMAA thresholds is a problem in its own right. The companion **Roth Conversion Optimizer** (sold separately, or as a bundle) is purpose-built for it.

To import a conversion plan

1. Design your strategy in the Roth Optimizer and click **Export plan** to save a `.roth-plan.json` file.
2. In the Retirement Planner, go to the Edit values tab.
3. Find the **Roth Conversion Plan Import** section and click **Choose .roth-plan.json file**.
4. The planner maps each year to the right phase and sets the per-phase Roth conversion fields automatically.

Only the Roth conversion fields are updated; everything else stays as it was. Then open the **Plan with AI** sidebar and ask it to stress-test the imported conversions against your ACA cliff and IRMAA exposure — it sees the new conversions in its plan context.



Import an optimized conversion calendar and the planner fills your per-phase Roth fields automatically.

Part 6 — Learn as you plan

The planner is also a teaching tool. Several overlapping systems explain retirement concepts at the moment they matter, using your actual numbers wherever possible.

Chapter 28 — ⓘ icons: click any field you don't understand

More than two dozen fields show a small ⓘ icon next to their label. Click it for a ~60-word plain-English explanation of what the field is, why it matters, and a common misconception. Most popovers include a **Read more in Glossary** link.

Examples

- ⓘ next to *Roth conversion* explains conversions and how they raise MAGI. ⓘ
- ⓘ next to *IRMAA threshold* explains the 2-year lookback and the cliff nature.
- ⓘ next to *ACA 400% FPL cliff* explains the rule and the Roth-below-the-cliff strategy.
- ⓘ next to *Equity gain %* explains the embedded-gain concept.

New in v7 — more (and more precise) popovers

- A **getting-started** popover that points new users to enter their monthly withdrawals first.
- A **distinct 22%-bracket** explainer — previously the 12% and 22% bracket icons opened the same text; now each explains its own ceiling.
- A **dedicated SS-claiming-age** popover — the SS claiming-age icon no longer opens the same FRA text as the birth-year field; each now has its own focused explanation.
- A **cash/bond volatility (σ)** popover explaining sensible Monte Carlo ranges for lower-risk buckets.

ⓘ Tax bracket position ×

⚠ **Phase 2:** you're in the 22% bracket. **\$10,796** of taxable income is taxed at the higher rate.

Your taxable income for this phase: **\$63,481** (after the standard deduction of \$16,830).

12% bracket ceiling: **\$52,685** · 22% bracket ceiling: **\$110,492** (inflation-adjusted to this phase).

Strategy: reduce 401k withdrawals or substitute Roth (which doesn't add to taxable income) to drop back into the 12% bracket.

🔥 Ask AI about this

[Read more in Glossary →](#)

▶ 12% tax bracket headroom · 2 min

▶ Roth conversion · 3 min

Click any ⓘ for a plain-English explanation — now with separate popovers for the 22% bracket and SS claiming age.

Chapter 29 — Clickable phase-card badges

Every colored badge on a phase card is clickable. Unlike the ⓘ icons (which explain a concept generically), badge explanations use the actual numbers from that phase:

- "You're \$4,200 below the 12% ceiling. You could withdraw an extra \$4,200 from your 401k or convert that much to Roth without bumping into 22%."
- "Your phase MAGI is \$58,000 — \$2,200 above the 400% FPL cliff at \$55,800. Either reduce 401k withdrawal by \$2,200 or substitute equivalent Roth."
- "Your 401k withdrawal of \$15,000 is \$4,300 below your estimated RMD of \$19,300. Increase to at least the RMD level to avoid the 25% penalty."

Badges are designed to convert insight into action — where appropriate, the explanation includes a specific dollar amount and direction.

 **Tax bracket position** 

 **Phase 2:** you're inside the 12% bracket with **\$23,952** of headroom before crossing into 22%.

Your taxable income for this phase: **\$24,906** (after the standard deduction of \$24,040).

12% bracket ceiling: **\$48,858** · 22% bracket ceiling: **\$104,166** (inflation-adjusted to this phase).

Strategy: use the headroom to do Roth conversions or take extra 401k withdrawals — taxed at only 12%.

[Read more in Glossary →](#) [▶ 12% tax bracket headroom · 2 min](#) [▶ Roth conversion · 3 min](#)

Badges explain your situation in your own numbers — with a specific action to take.

Chapter 30 — Concepts-to-Learn panel

On the Overview tab, a collapsible card titled *Concepts to learn* surfaces 3–4 concepts most relevant to your current plan. The selection is driven by 10+ rules that look at your specific numbers — for example:

- You have a Roth balance but no scheduled conversions → a *Roth conversions in low-bracket years* card.
- You're within 15% of the ACA cliff in a pre-Medicare phase → a *Managing the ACA cliff with Roth* card.
- Your plan ends past 73 with a large 401k and no RMD-aware withdrawals → an *RMD planning before 73* card.
- Your spouse age gap is 3+ years → a *Spouse claim-age timing* card.
- Your later-phase withdrawals are as high as (or higher than) your early ones → a *Model the retirement spending smile* card (see the Glossary entry in Chapter 31).

The card collapses to a thin status bar when you want to focus on numbers, and your choice persists. Each tile opens the relevant ⓘ explanation or glossary entry.

The screenshot shows a retirement planning interface. At the top, there's a summary bar with metrics: AVG NET / MO (\$4,598 nom, \$3,843 real), EST. TOTAL AGE 80 (USD) (\$1,146,098), PHASE 5 PROJECTION: Age 72-80, Extend to age 90, and Total tax to 80: \$62,563. Below this, starting balances are listed: STARTING 401K (USD) \$400,000, STARTING CASH (USD) \$50,000, and STARTING EQUITY (USD) \$80,000. The main section is titled 'Concepts to learn — based on your plan' and contains four personalized suggestions:

- Start here: add your monthly withdrawals**
Add your account balances in Edit values or directly on the dashboard, then enter how much to withdraw from each balance every month during each phase — these build your total monthly income projection.
- You could qualify for ACA Silver CSR**
Pulling MAGI in Phase 2 below 250% FPL would unlock Cost-Sharing Reductions — much lower deductibles. Often saves \$4–8k/yr.
- You have unused 12% bracket room**
In Phase 2 you have about \$23,952 of headroom below the 12% bracket ceiling. Roth conversions or 401k withdrawals up to that amount are taxed at only 12%.
- RMD shortfall in your plan**
In Phase 5, your 401k withdrawal is below the IRS Required Minimum. Penalty: 25% of the shortfall.

Personalized learning prompts, chosen from your own plan's situation.

Chapter 31 — Glossary and the 9-step Feature Tour

Searchable A–Z Glossary

The Glossary tab in the Help modal contains 20+ entries — every retirement term used in the planner, defined with typical values, common misconceptions and related concepts. Type to filter; entries match on term, synonym and body text.

New in v7 — the retirement spending smile (go-go / slow-go / no-go)

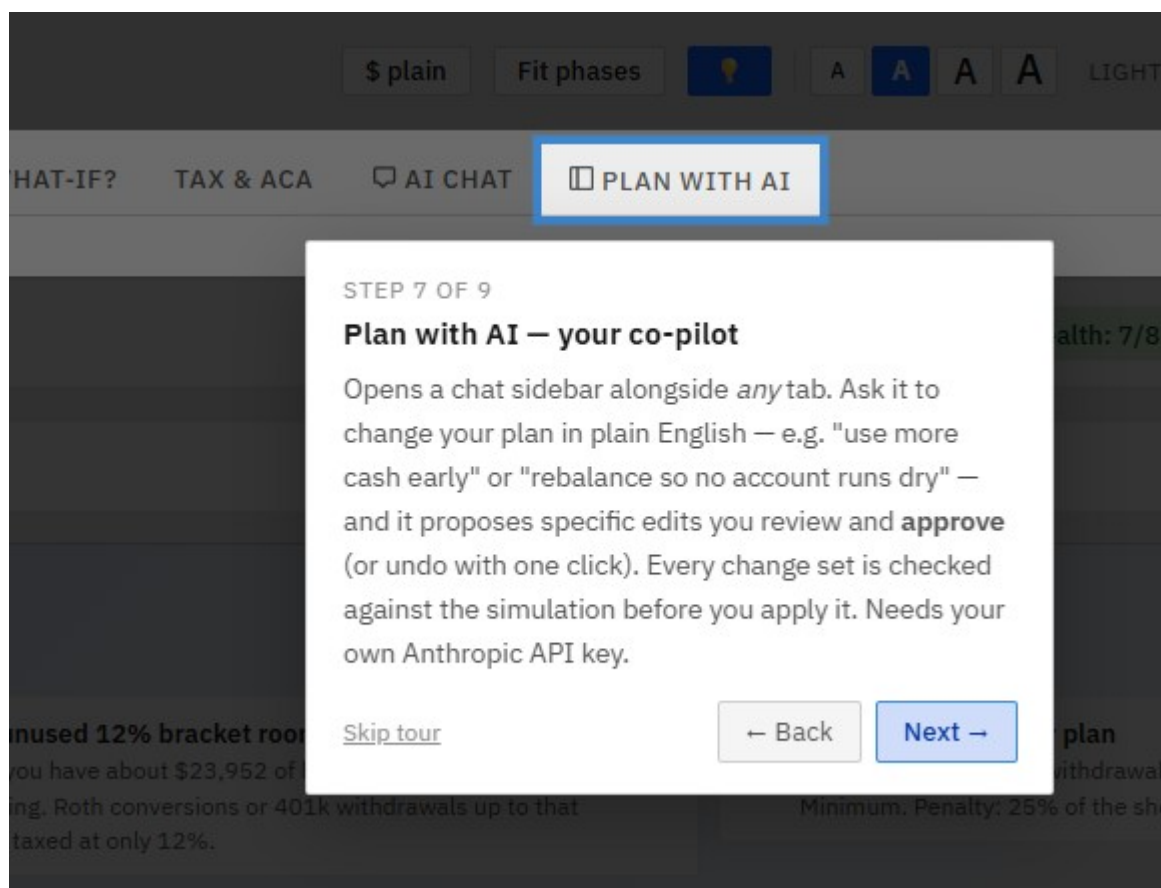
Studies of actual retiree spending find it rarely stays flat — it tends to trace a **smile**: the active **go-go** years (travel, hobbies, helping family) cost more; the **slow-go** years settle down; the late **no-go** years can rise again on health and care costs. **How to model it here:** set higher withdrawals in earlier phases and lower ones in the middle, then consider a bump in Phase 5 — or a one-time *money-out* expense (Chapter 8) — for late-life care. Because the planner uses fixed *nominal* withdrawals, even keeping them level lets your *real* spending drift down; shaping the curve deliberately makes the plan match how you'll actually spend. Reachable from the Concepts-to-learn panel (Chapter 30) and this Glossary.

Guided Feature Tour

From the Help modal, click **Take the Feature Tour** for a guided overlay that highlights each major area with a short caption.

New in v7 — expanded to 9 stops

The tour now covers nine areas, adding stops for the new **Plan with AI** co-pilot and the **Report** button alongside the original seven (tab bar, Plan Health badge, Replan button, summary cards, Concepts to learn, phase cards, Help). The tour correctly lifts the sticky header out of the way so highlighted targets are never hidden behind the pinned bars.



The 9-step tour now introduces the Plan with AI co-pilot and the Report button.

Chapter 32 — Tax & ACA notes tab

The **Tax & ACA** tab is the technical reference for every calculation in the planner — tax bracket mechanics, IRMAA, SS provisional income, ACA cliff rules, UK residence and the Foreign Tax Credit, equity treatment, replan behavior and more. If a number on the dashboard surprises you, this is the first place to look. A sticky table of contents on the left lists 23 sections with jump-links; mobile collapses it to a single column. Sentences carry small *AI-corrected* badges where you've applied audit corrections (Chapter 22).

AI Retirement Income Planner v7

Custom rates

Export JSON

Import JSON

Create Report

Reset defaults

?

Scenario: Age 59.5, Downsize US Condo Phase 1 / 5-Phase Retirement Income Plan - ACA - CSR - Tax - Inflation-Adjusted Net Income - Lump Sums - Equities

USD

GBP

CAD

AUD

PHP

THB

Edit custom rates

\$ plain

Fit phases

⬆

A

A

A

LIGHT

STYLE

Pro dashboard

OVERVIEW

EDIT VALUES

CASH LUMP SUMS

BALANCE

SCENARIOS

STRESS TEST

SS OPTIMIZER

DRAWDOWN

ANNUITIES

WHAT-IF?

TAX & ACA

AI CHAT

PLAN WITH AI

JUMP TO TOPIC

Currency

Inflation & real income

UK Triple Lock & SS COLA

Tax brackets

IRMAA

ACA & FPL

Foreign residence

UK residence (FTC)

SS provisional income

Equities

Part-time work

Replan from today

Save / Export

MFJ filing

Roth IRA

Roth conversions

Rental income

RMD estimate

Custom phase boundaries

Canada CPP & OAS

Australia Super

Stress test

Monte Carlo

HOW CALCULATIONS WORK

Reference notes for every calculation in the dashboard. Use the table of contents on the left to jump straight to a topic. For plain-English definitions of individual concepts, open the **Glossary** tab in the Help modal.

Currency: All calculations run in USD; displayed values convert using live or manual FX rates.

Inflation & real income: Real = nominal + (1+inflation)ⁿyears. Phase mid-point used for deflation. [Real vs nominal - 2 min](#) [Inflation - 2 min](#)

UK Triple Lock & SS COLA: Pensions grow once per year in the month-by-month simulation. SS grows annually from age 62; UKP from age 67. [SS COLA - 90 sec](#) [UK Triple Lock - 2 min](#)

Tax brackets: Fully configurable base-year values. 10% up to your configured ceiling - 12% to your ceiling - 22% to your ceiling - 24% above. All bracket boundaries, deductions, and FPL thresholds are automatically inflated forward to each phase's midpoint using your general inflation rate, approximating annual IRS/HHS adjustments. The inflated values are shown on each phase card. [12% bracket - 2 min](#) [Taxable income - 90 sec](#)

IRMAA: If MAGI exceeds the IRMAA threshold (configurable in Edit tab), a warning is shown on the phase card. The threshold is inflation-adjusted per phase like other tax parameters. [IRMAA - 3 min](#) [MAGI - 3 min](#)

ACA & FPL: Prior-year FPL values are used for ACA eligibility, consistent with marketplace guidelines. FPL thresholds are inflation-adjusted forward in calculations. [ACA cliff - 3 min](#) [ACA floor - 2 min](#) [FPL - 90 sec](#) [ACA subsidy - 2 min](#)

Foreign residence mode: When a non-USD currency is selected (GBP, PHP, THB), ACA premiums, Medicare Part B/D, ACA subsidy/CSR eligibility, and IRMAA checks are excluded from calculations and shown as N/A on the dashboard. This reflects a scenario where the retiree is living outside the United States.

UK residence mode (GBP only): When GBP is selected and UK Residence is active, UK income tax is calculated on 401k withdrawals + UK State Pension + equity gains + part-time income. US Social Security is NOT taxed in the UK under the US-UK treaty (Article 17 — taxed only in the US). US federal tax is reduced by a Foreign Tax Credit equal to the UK tax paid (capped at US tax liability), preventing double taxation. The effective tax paid is the higher of the two countries' tax on the overlapping income. UK tax brackets (personal allowance, basic/higher/additional rates) are configurable in the Edit tab and inflation-adjusted per phase. [FTC - 3 min](#) [US-UK Treaty \(SS\) - 90 sec](#) [UK allowance & bands - 2 min](#)

SS provisional income: Other income + part-time income + 50% of SS. Under \$25k — 0% taxable. \$25–34k — 50%. Above \$34k — 85%. [SS Provisional - 2 min](#)

Equities: Equity withdrawals are treated as post-tax capital returns (not included in ordinary income tax). They are included in MAGI for ACA calculations. The equity balance compounds at your specified annual rate of return. [Equity gain - 2 min](#) [LTCG - 2 min](#) [Qualified dividends - 90 sec](#)

Part-time work: Annual income entered for any phase is treated as ordinary income (included in gross income for tax purposes). It counts toward your tax brackets, MAGI (for ACA/IRMAA), and SS provisional income just like a 401k withdrawal.

ACA cliff (2026): 400% FPL cliff returned Jan 1 2026. [ACA cliff - 3 min](#)

Replan from today: Use the replan button to fast-forward the dashboard to your current age. Past phases are removed, the current phase is trimmed, and calculations restart from your actual balances. When replanning, enter your current SS and UKP payment amounts — COLA and triple lock growth will be calculated forward from that point, not from the original base ages. [Replan - 90 sec](#)

The technical reference behind every number, with a jump-link table of contents.

Part 7 — Settings and maintenance

The last part covers the planner's personalization options, data export/import, and the long-term workflow for keeping your planner current as tax law evolves.

Chapter 33 — Text size, dark mode, themes & the polished header

Text size & dark mode

Buttons in the toolbar (A-A-A-A) set the planner's text size; the moon/sun button toggles light/dark. Both persist across sessions. Dark mode reduces eye strain in low light; light mode prints better.

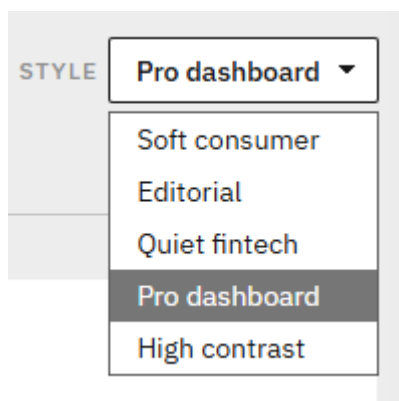
Visual styles (themes)

The **Style** picker offers five visual styles you can switch between at any time:

- **Soft consumer** (default) — friendly, rounded.
- **Editorial** — magazine-like.
- **Quiet fintech** — restrained and data-forward.
- **Pro dashboard** v7 — denser, tool-like.
- **High contrast** v7 — an accessibility theme for maximum legibility.

New in v7 — a calmer, clearer layout

- **Workflow tab order:** tabs read left-to-right as a workflow — inputs (Overview, Confidence, Edit values, Lump sums), then visualize (Balance, Scenarios, Stress test), then strategy (SS Optimizer, Drawdown, Annuities, What-if?), then reference (Tax & ACA), then AI Chat and Plan with AI.
- **Polished sticky header:** as you scroll, the tab bar lifts with a soft shadow and a frosted background, and a slim marker slides under the active tab. (Motion is automatically disabled if your system prefers reduced motion.)
- **Grouped Overview toolbars:** the controls along the top are grouped by purpose with thin dividers — dashboard-view controls separated from appearance controls, and data/output actions separated from Reset and Help.



Five visual styles — including the new Pro dashboard and High contrast themes.

Grouped toolbars and a polished sticky header keep the controls calm and oriented.

Chapter 34 — Saving, exporting, importing

Auto-save

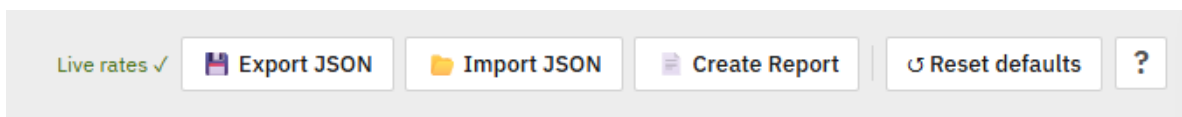
Every change is automatically saved to your browser's local storage. Re-open in the same browser and your plan is still there — nothing to remember to click.

Export & import plan as JSON

Click **Export JSON** to download a `.json` file containing your complete plan — every balance, withdrawal, tax parameter, lump sum, acknowledged check, AI correction and annuity. Click **Import JSON** to load a previously-exported file (it replaces what's in your browser). Useful for backups, switching scenarios, or moving between machines.

Privacy reminder

Your plan stays on your computer unless you export it or send it to your AI provider. Auto-save writes only to your browser's local storage — not to any cloud service.



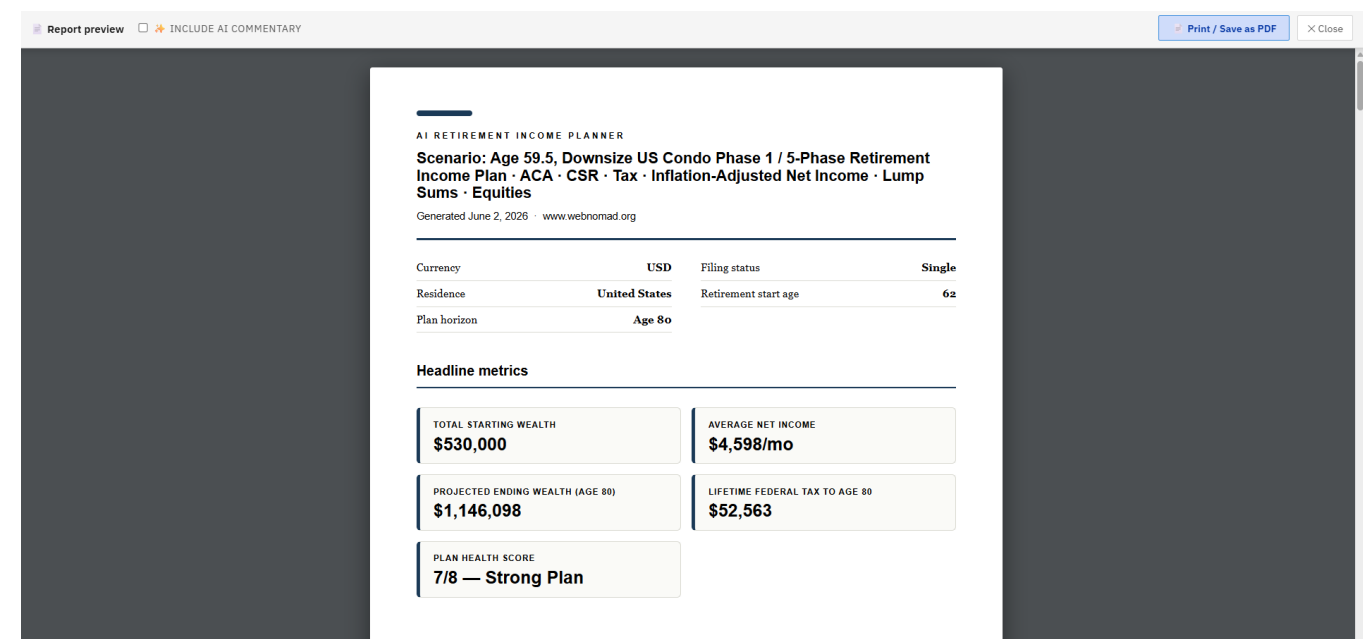
Export a complete JSON backup, or import one to restore or share a plan.

Chapter 35 — PDF export

Create a clean, multi-page report with the **Report** button (grouped with Export/Import on the Overview title row).

- 1. Click **Report**.
- 2. Tick the sections you want: Include notes, Include calculations, Include AI commentary. All three start off each time you open the planner, and the commentary tick needs an API key.
- 3. Review the preview — AI prose appears in editable blocks you can adjust or skip.
- 4. Print to PDF (Save as PDF) and choose where to keep the file.

See Chapter 23 for the full contents — including the new asset-allocation, plan-resilience and portfolio-charts pages, and the blank-page fix.



Choose AI commentary or rule-based only — then review and edit before saving the PDF.

Chapter 36 — Keeping your planner current (the long-term workflow)

Most planners go stale the moment tax law moves. This one has four built-in mechanisms. Use them as a quarterly or annual maintenance routine.

The four mechanisms

Tier 1 — Auto-updating popovers. When you run **Fetch current rates** (Chapter 21), five info popovers automatically reflect the new numbers — IRMAA threshold, ACA cliff, ACA CSR band, 12% bracket, standard deduction.

Tier 2 — AI audit of structural claims. Click **Audit help content** (Chapter 22) to verify 15 factual claims against today's law. Read-only by default.

Tier 3 — Verified watermark. Every Help pane shows when the help text was last audited (or "never audited — content baseline"). After 18 months the watermark turns amber.

Tier 4 — One-click prose corrections. When the audit finds a changed claim, click **Apply this update everywhere** to substitute the corrected prose into every place that fact appears, with an *AI-corrected* badge. Click **Revert** to undo.

Recommended quarterly routine

1) Fetch current rates and review the diff. 2) Audit help content and apply/revert updates. 3) Check the verified-watermark date in the Help modal. 4) Export your plan as JSON, saved with today's date — a permanent record of your plan as verified.

Keep the help content evergreen

Tax laws change. The **Audit help content** button asks Claude to verify the content; it never modifies your plan or the help content; it just produces a stamp.



Every Help pane shows when its content was last verified against current law.

Chapter 37 — Troubleshooting and FAQ

Q: I open the file and see blank/strange formatting.

Use a modern browser (Chrome, Edge, Firefox or Safari from the last 2–3 years). Internet Explorer and very old Edge aren't supported.

Q: My plan disappeared after a Windows/browser update.

Clearing your browser's site data wipes the planner's local storage. Always export a JSON backup before a major OS or browser update, and re-import after.

Q: I'm not a US citizen — does the planner still work for me?

Yes. Choose **Non-US — no US tax** under Edit values → Income. US federal tax, IRMAA, ACA and SS-provisional calculations zero out. UK, Canadian and Australian regimes still work.

Q: The Plan with AI sidebar keeps suggesting changes that run me out of money — why does it retry?

When you ask for a plan that doesn't deplete, the co-pilot tests each suggestion in the simulator and, if it still depletes, asks the AI to refine — up to three attempts. If it can't find a solvent plan, that's a signal your targets may be infeasible without bigger changes (later SS, lower spending, more conversions). See Chapter 20.

Q: Is my data — or my API key — uploaded anywhere?

Plan data stays in your browser's local storage. The planner fetches exchange rates as it opens and checks daily for updates, sending no plan data. AI features send a plan summary to your chosen provider only when used; your key is stored only in your browser. Full list: our website's Technical page.

Q: Can I share it with my financial advisor?

Yes — export your plan as JSON or PDF. The `.html` file license is single-user; for redistribution, contact WebNomad Studio.

Q: Is this financial advice?

No. The planner is an educational tool. Always verify your assumptions and consult a qualified professional before acting on any output.

US TAX OBLIGATION

US taxpayer

Non-US — no US tax

Toggle **Non-US** if you're not a US person under IRS rules (no US citizenship, no green card, no substantial presence).
When off, all US federal tax, IRMAA, and ACA calculations are skipped — UK / Canadian / Australian tax regimes still apply if their flags are set.

Not a US person? One toggle removes all US-specific tax and healthcare modeling.

Appendix A — Keyboard shortcuts

The planner uses standard browser shortcuts:

- **Ctrl/Cmd + P** — open the print / PDF export dialog.
- **Ctrl/Cmd + F** — find text on the current page (useful in the Tax & ACA notes tab). **Esc**
- — close any open popover, modal, or the Plan with AI sidebar overlay on mobile. **Tab**
- **/ Shift+Tab** — navigate between input fields on the Edit values tab.

Appendix B — JSON export field reference

The plan JSON export contains a handful of top-level keys:

Key	Meaning
s	The full state object — every input value, including per-phase objects p1–p5 (now with a part-time income field in each phase) and state-tax settings.
lumps	Lump-sum events: id, amtUSD, phase, label, dir, accounts, ticks.
p5EndAge	The age your plan runs to (default 90).
activeCurrency	Your active residency currency (USD / GBP / CAD / AUD / PHP / THB / EUR).
subtitle	Your scenario title (shown on the dashboard and PDF cover).
annuityPlanState	Records an applied annuity's premium, per-phase income additions and funding deductions for clean removal.
learnDismissed	Concepts-to-Learn entries you've dismissed.
phAcknowledged	Plan Health checks you've acknowledged (key, optional reason, timestamp).
helpOverrides	AI-applied audit corrections that survive reload and travel with the plan.
v / date	Format version and export timestamp — used for compatibility detection on import.

Note on AI settings

Your API key and the two chat histories (the AI Chat tab and the Plan with AI sidebar) are stored separately in browser local storage, *not* in the exported plan JSON — so sharing a plan never shares your key or your conversations.

Appendix C — Where to learn more

YouTube playlist

The WebNomad Studio YouTube playlist includes walkthrough videos covering the dashboard, first scenario setup, the Tax & ACA notes deep-dive, comparing scenarios, the Philippines expat scenario, the Roth conversion workflow, drawdown strategies, stress testing, and the AI advisor in practice. Search "*WebNomad Studio retirement planner*" on YouTube.

Where to buy

The Retirement Income Planner (this product) and the Roth Conversion Optimizer (Chapter 27) are sold directly at airetirementincomeplanner.com. A bundle of both is available at a discount.

WebNomad Studio

Website: airetirementincomeplanner.com · Email: dev@webnomad.org. Refined across seven major versions based on real user feedback from the retirement-planning community.

Retirement Income Planner — Version 7 User Guide · WebNomad Studio · webnomad.org · Published June 2026, updated October 2026. This guide supersedes the v6 User Guide.